ACHIEVE!

DISC INSIDE

IWB FRIENDLY

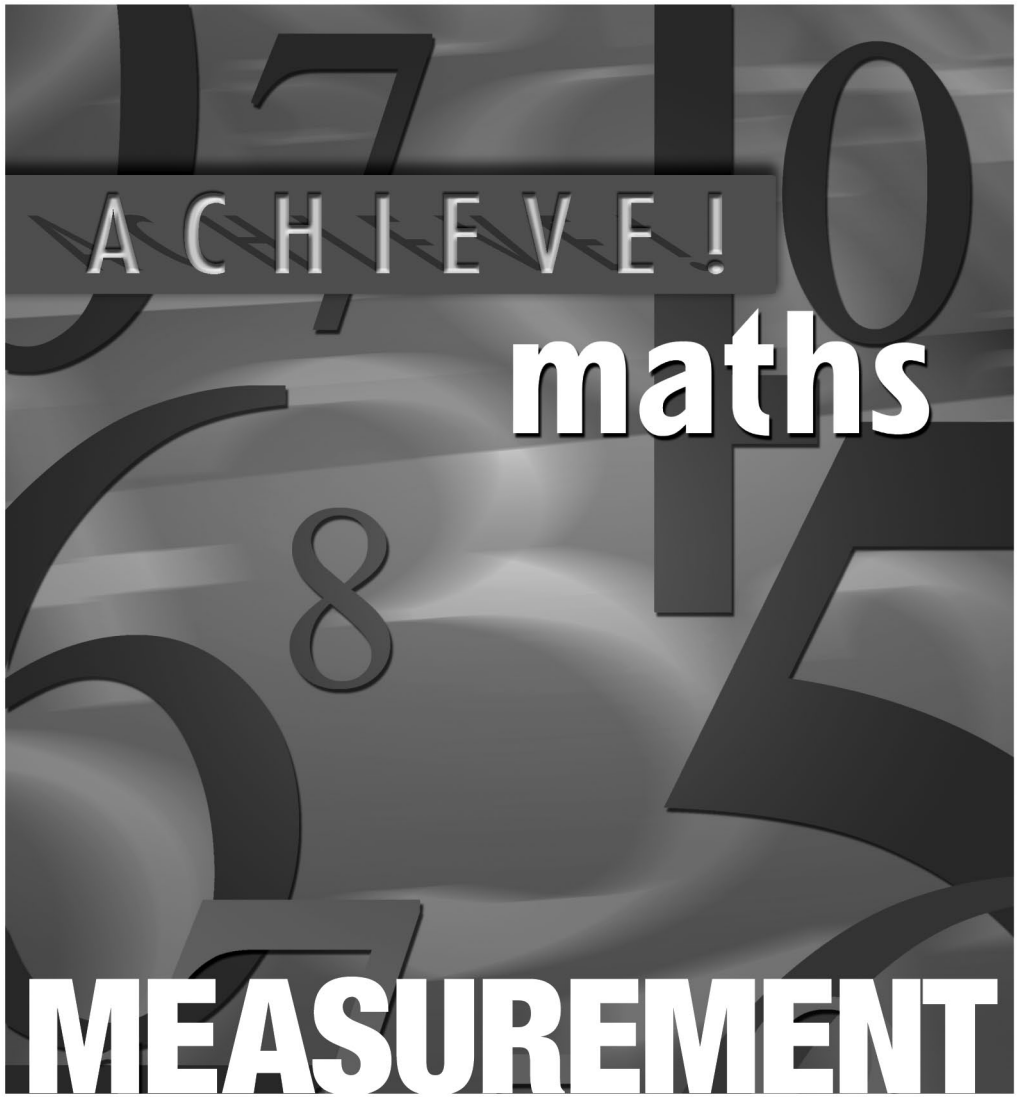
maths

MEASUREMENT

Paul Steenson

INSTANT LESSONS

DISC INSIDE



Paul Steenson

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Paul Steenson has more than thirty years' experience teaching Mathematics in secondary schools in Australia. He has also worked as a test development officer and been an assessor of Mathematics papers for the NSW HSC examination. Paul served as a member of the NSW Mathematics Syllabus Committee and has been a marker of both School Certificate and HSC examination papers. He has been a regular presenter of curriculum workshops and teacher development activities. He has also been a co-author of textbooks and written a number of titles in the Instant Lessons series.

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Introduction

The *Achieve!* series of Maths titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students. It is aimed at students who, because of poor literacy skills, struggle to read and comprehend regular classroom materials. Some activities are more challenging than others. The activity sheets are also suitable as supplementary material for regular classes. They may also provide useful revision in preparation for state-wide or national numeracy testing.

This book is divided into five units, each of which focuses on a topic or set of skills to do with measurement. The CD contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other activity sheets. The activity sheets are provided as interactive PDFs, which are ideal for use on interactive whiteboards (IWBs) and laptops. Teachers should note that students can save their answers and any other changes they make. For greater accuracy, questions that require students to measure, circle or draw should be completed on hard copies or on the IWB using the IWB tools.

Note that photocopying and printing copies from the CD may change the length of some lines and the size of diagrams. When students are asked to measure lengths, it is recommended that the teacher checks the answers using the same copy that is given to students.

Teachers can use the material in a variety of ways. For example, a unit could be taught in sequential lessons with a whole class or with students in groups of varying sizes. Alternatively, one or more activity sheets can be used as support material, for individual homework, assessment or revision. Teachers will need to select carefully if individual activity sheets are used as stand-alone tasks, as some activities build on knowledge gained earlier in the unit.

The Teacher's notes provide valuable guidance about using each unit, and contain the following elements.

Objectives – These are the key skills and knowledge learnt through using the activity sheets.

Prior knowledge – This refers to the skills and knowledge that are required for students to complete the tasks.

Background – This section provides a range of different types of useful information to help teach this unit. It can include explanations of key concepts and terminology or further background notes about the subject matter.

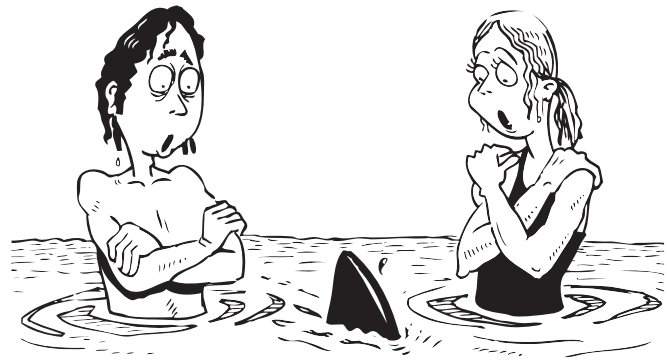
Starter activity – Warm-up activities are provided to put the content into proper context and to gain student interest and attention.

Activity sheets and answers – A summary of the content of each activity sheet is provided, along with teaching tips and selected answers to assist with time-efficient marking.

Extension ideas – Additional activities or discussion ideas are suggested to recap on the main skills, knowledge or terminology taught, as well as ways to extend, assess or consolidate student understanding.

Curriculum links – The activity sheets will enable students to achieve outcomes required for the following State levels or stages:

- New South Wales – Stages 4 and 5
- Victoria – Levels 5 and 6
- Queensland – Levels 4, 5 and 6



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Teacher's notes

OBJECTIVES

- Understand the concepts of length and perimeter.
- Measure and record a length in centimetres or millimetres.
- Convert between millimetres, centimetres, metres and kilometres to compare lengths or distances.
- Find the perimeter of a plane shape by measurement.
- Calculate the perimeter of various plane shapes, including rectangles, squares and triangles.

PRIOR KNOWLEDGE

Students should be able to measure the length of an interval using a ruler marked in centimetres and millimetres. They should be able to perform the basic operations (+, –, ×, ÷) with whole numbers and decimals using mental arithmetic or a calculator. They should also be able to recall and apply the properties of a rectangle, square, parallelogram and rhombus.

BACKGROUND

The term ***Perimeter*** comes from the Greek word that means *to measure around the outside*.

Students entering any of the trades related to the building industry will need to confidently use millimetres to measure lengths.

STARTER ACTIVITY

Ask students to estimate various lengths in the classroom then use rulers, tape measures or trundle wheels to measure these lengths. Discuss how various lengths and distances are recorded.

NOTE: Photocopying and printing copies from the CD may change the length of some lines and the size of diagrams. When students are asked to measure lengths, it is recommended that the teacher checks the answers using the same copy that is given to students.

ACTIVITY SHEETS

1.1 Measuring lengths

This activity sheet reminds students of the metric units for measuring length. They are first asked to measure the length of a number of intervals to the nearest centimetre. In question 3 they must measure to the nearest millimetre. In question 4, they are required to select the most appropriate unit to use in measuring lengths and distances. A ruler is required to complete these activities.

Answers

2 **a** 4 cm **b** 6 cm **c** 3 cm **d** 5 cm **e** 7 cm **f** 4 cm **g** 10 cm **h** 12 cm

3 **a** 45 mm **b** 55 mm **c** 36 mm **d** 47 mm **e** 61 mm **f** 53 mm

4 **a** m **b** km **c** mm **d** m **e** mm **f** m **g** m **h** mm **i** mm **j** m **k** km **l** mm **m** m **n** m **o** km **p** mm **q** mm **r** mm **s** m **t** km

1.2 Units of length

This activity sheet develops students' skills in converting between different units for measuring length. Encourage them to learn the conversion rules at the top of each page

Answers

1 **a** 12 000 m **b** 9 500 m **c** 1 700 m **d** 800 m **e** 6 120 m **f** 16 500 m

2 **a** 14 000 mm **b** 8 200 mm **c** 11 300 mm **d** 700 mm **e** 1 280 mm **f** 5 630 mm

3 **a** 800 cm **b** 1 400 cm **c** 640 cm **d** 1 130 cm **e** 20 cm **f** 225 cm

4 **a** 70 mm **b** 240 mm **c** 87 mm **d** 9 mm **e** 113 mm **f** 1 500 mm

5 **a** 2 600 m **b** 6 350 mm **c** 460 mm **d** 3 480 m **e** 62 cm **f** 4 070 mm

6 **a** 4 km **b** 7.2 km **c** 15.6 km **d** 8.35 km **e** 0.75 km **f** 2.04 km

7 **a** 3 m **b** 15 m **c** 4.9 m **d** 26.3 m **e** 0.87 m **f** 4.65 m

8 **a** 6 m **b** 12 m **c** 6.7 m **d** 19.6 m **e** 1.75 m **f** 0.95 m

9 **a** 4 cm **b** 80 cm **c** 600 cm **d** 35 cm **e** 6.5 cm **f** 2.45 cm

10 **a** 5 cm **b** 5.4 km **c** 8.5 m **d** 0.78 m **e** 4.56 m **f** 14.75 km

11 **a** 7 km **b** 5.7 m **c** 175 mm

12 **a** 2.5 km **b** 0.7 m **c** 6 510 cm

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Teacher's notes

CONTINUED

1.3 Measuring perimeters

This activity sheet introduces the concept of the perimeter of a plane shape as the distance around the outside edge. Students need to measure the length of each side of a shape and then add these measurements to find the perimeter. A ruler is required to complete these activities.

Answers

1 **a** 12 cm **b** 10 cm **c** 12 cm **d** 12 cm
e 12 cm **f** 12 cm

2 **a** 15 cm **b** 16 cm **c** 20 cm **d** 20 cm

3 **a** 146 mm **b** 168 mm **c** 170 mm
d 160 mm

1.4 Calculating perimeters A

In this activity sheet, students are given the length of each side of a plane shape. These diagrams are not drawn to scale. Encourage students to write down the addition they need to perform as well as the perimeter. In the last question, they are given the perimeter of the shape and asked to use subtraction to find a missing length.

Answers

1 **a** 31 cm **b** 45 cm **c** 80 cm **d** 68 cm
e 126 cm **f** 120 cm **g** 180 mm
h 260 mm **i** 230 mm

2 **a** 46 cm **b** 66 cm **c** 48 cm **d** 280 mm
e 310 mm **f** 380 mm

3 **a** 35 cm **b** 28 cm **c** 23 cm

1.5 Calculating perimeters B

In this activity sheet, students are given sufficient data to calculate the perimeter of common plane shapes – rectangles, squares, parallelograms and rhombuses. They need to use the properties of the shape to calculate the perimeter.

Answers

1 **a** 104 cm **b** 92 cm **c** 190 cm **d** 178 mm
e 200 mm **f** 210 mm

2 **a** 68 cm **b** 112 cm **c** 64 cm **d** 144 mm
e 216 mm **f** 292 mm

EXTENSION IDEAS

Ask students to describe how they would estimate or measure a length or distance. Guide their responses by using specific lengths such as those listed in Activity sheet 1.1 Measuring lengths question 4.

Then discuss possible reasons why two students may obtain different measures for the same length, distance or perimeter. Encourage students to consider the effects of rounding off, choice of units and human error.

Have students work in small groups to find the perimeter of a large area such as the school sports ground or, for homework, their house or flat.

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1.1 Measuring lengths

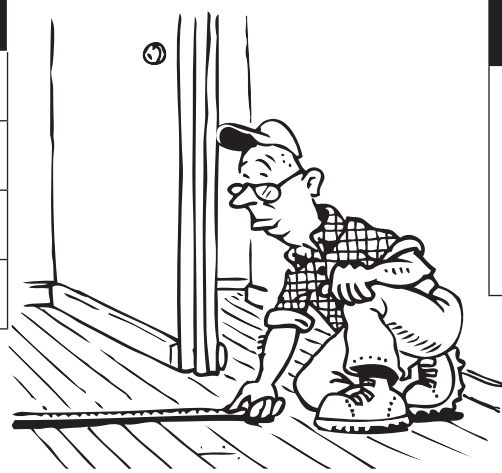
We use a metric system of units to measure lengths and distances.
The basic unit of length is the **metre**.
One metre is about the length of a step or pace.

1 How wide is your classroom from wall to wall?

a Pace it out and write down your estimate._____

b Now check your estimate using a tape measure._____

Name	Symbol
kilometre	km
metre	m
centimetre	cm
millimetre	mm



Comparing units
1 kilometre = 1 000 metres
1 metre = 100 centimetres
1 centimetre = 10 millimetres
1 metre = 1 000 millimetres

2 Use a ruler to measure each interval to the **nearest centimetre**.
Write your measurements in the spaces provided.

a

.....

b

.....

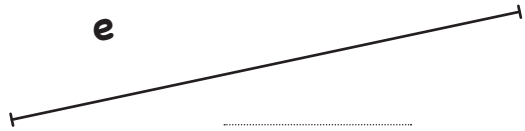
c

.....

d

.....

e



.....

f

.....

g

.....

h

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1.1 Measuring lengths

CONTINUED

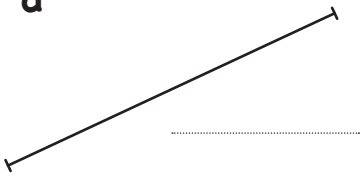
3 Use a ruler to measure each interval to the **nearest millimetre**. Write your measurements in the spaces provided.

a

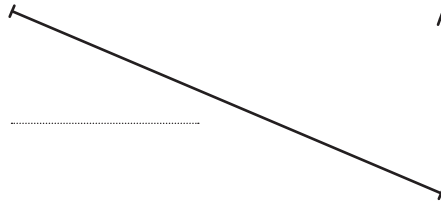
b

c

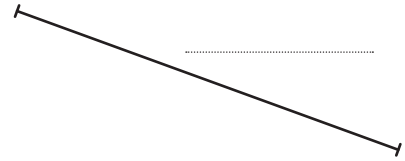
d



e



f



4 Which unit – **kilometre (km)**, **metre (m)** or **millimetre (mm)** – would you use to measure each of the following? Write the unit abbreviations.

a the length of the classroom __*m*__

b the distance from Sydney to Melbourne _____

c the width of your hand _____

d the length of a hockey field _____

e the height of a can of drink _____

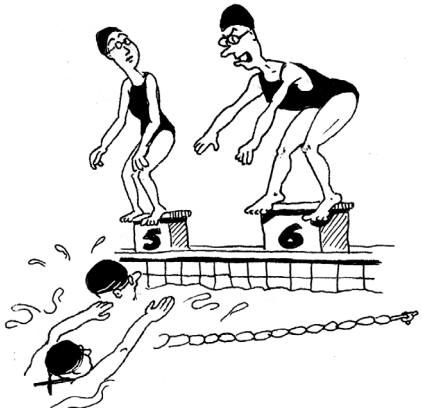
f the height of a tree _____

g the length of a cricket pitch _____

h the length of your shoe _____

i the width of a brick _____

j the width of a swimming pool _____



k the length of the Equator _____

l the diameter of a DVD _____

m the height of a flagpole _____

n the length of a tennis court _____

o the distance from home to school _____

p the width of a doorway _____

q the length of a finger _____

r the width of your desk _____

s the width of a street _____

t the length of the Murray River _____

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1.2 Units of length

All measurements on building plans are written in millimetres.

A room is 4.3 metres long.

What length is written on the plan?

10 millimetres (mm)	= 1 centimetre (cm)
1 000 millimetres (mm)	= 1 metre (m)
100 centimetres (cm)	= 1 metre (m)
1 000 metres (m)	= 1 kilometre (km)

Converting to smaller units:

- To convert ***kilometres*** to ***metres***, **multiply** by 1 000.
- To convert ***metres*** to ***millimetres***, **multiply** by 1 000.
- To convert ***metres*** to ***centimetres***, **multiply** by 100.
- To convert ***centimetres*** to ***millimetres***, **multiply** by 10.

EXAMPLES

Convert 5 km to metres.

5 × 1 000 = 5 000 m

Convert 4.3 m to millimetres.

4.3 × 1 000 = 4 300 mm

A quick way to **multiply** by **1 000**: ***Move the decimal point 3 places to the*** **right**.

1 Convert to metres (m).

a 12 km = _____ **b** 9.5 km = _____ **c** 1.7 km = _____

d 0.8 km = _____ **e** 6.12 km = _____ **f** 16.5 km = _____

2 Convert to millimetres (mm).

a 14 m = _____ **b** 8.2 m = _____ **c** 11.3 m = _____

d 0.7 m = _____ **e** 1.28 m = _____ **f** 5.63 m = _____

3 Convert to centimetres (cm). *Hint:* multiply by 100.

a 8 m = _____ **b** 14 m = _____ **c** 6.4 m = _____

d 11.3 m = _____ **e** 0.2 m = _____ **f** 2.25 m = _____

4 Convert to millimetres (mm). *Hint:* multiply by 10.

a 7 cm = _____ **b** 24 cm = _____ **c** 8.7 cm = _____

d 0.9 cm = _____ **e** 11.3 cm = _____ **f** 150 cm = _____

5 Complete these conversions.

a 2.6 km = _____ m **b** 6.35 m = _____ mm **c** 46 cm = _____ mm

d 3.48 km = _____ m **e** 0.62 m = _____ cm **f** 4.07 m = _____ mm

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1.2 Units of length

CONTINUED

Converting to larger units:

- To convert ***metres*** to ***kilometres***, **divide** by 1 000.
- To convert ***millimetres*** to ***metres***, **divide** by 1 000.
- To convert ***centimetres*** to ***metres***, **divide** by 100.
- To convert ***millimetres*** to ***centimetres***, **divide** by 10.

EXAMPLES

Convert 13 000 m to kilometres.
13 000 ÷ 1 000 = 13 km

Convert 2 500 mm to metres.
2 500 ÷ 1 000 = 2.5 m

A quick way to **divide** by **1 000**: ***Move the decimal point 3 places to the*** **left**.

6 Convert to kilometres (km).

a 4 000 m = _____ **b** 7 200 m = _____ **c** 15 600 m = _____

d 8 350 m = _____ **e** 750 m = _____ **f** 2 040 m = _____

7 Convert to metres (m).

a 3 000 mm = _____ **b** 15 000 mm = _____ **c** 4 900 mm = _____

d 26 300 mm = _____ **e** 870 mm = _____ **f** 4 650 mm = _____

8 Convert to metres (m). *Hint:* divide by 100.

a 600 cm = _____ **b** 1 200 cm = _____ **c** 670 cm = _____

d 1 960 cm = _____ **e** 175 cm = _____ **f** 95 cm = _____

9 Convert to centimetres (cm). *Hint:* divide by 10.

a 40 mm = _____ **b** 800 mm = _____ **c** 6 000 mm = _____

d 350 mm = _____ **e** 65 mm = _____ **f** 245 mm = _____

10 Complete these conversions.

a 50 mm = _____ cm **b** 5 400 m = _____ km **c** 850 cm = _____ m

d 780 mm = _____ m **e** 4 560 mm = _____ m **f** 14 750 m = _____ km

11 For each pair of lengths, ***circle*** the **longer** length.

a 7 km or 6 950 m **b** 575 mm or 5.7 m **c** 17.3 cm or 175 mm

12 For each pair of lengths, ***circle*** the **shorter** length.

a 2.5 km or 24 500 m **b** 750 mm or 0.7 m **c** 650 m or 6 510 cm

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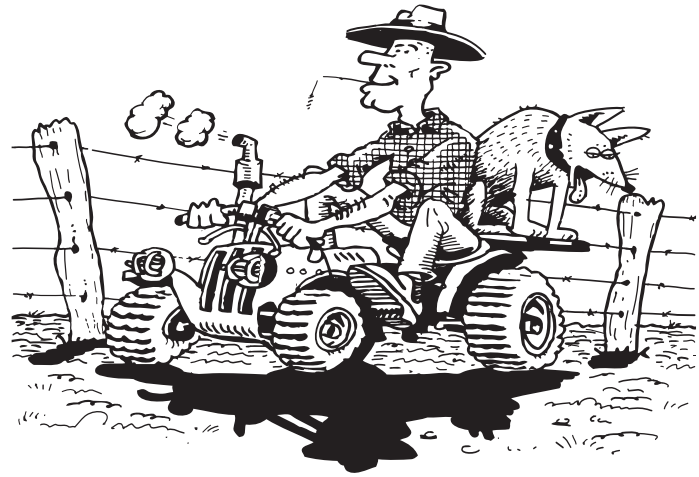
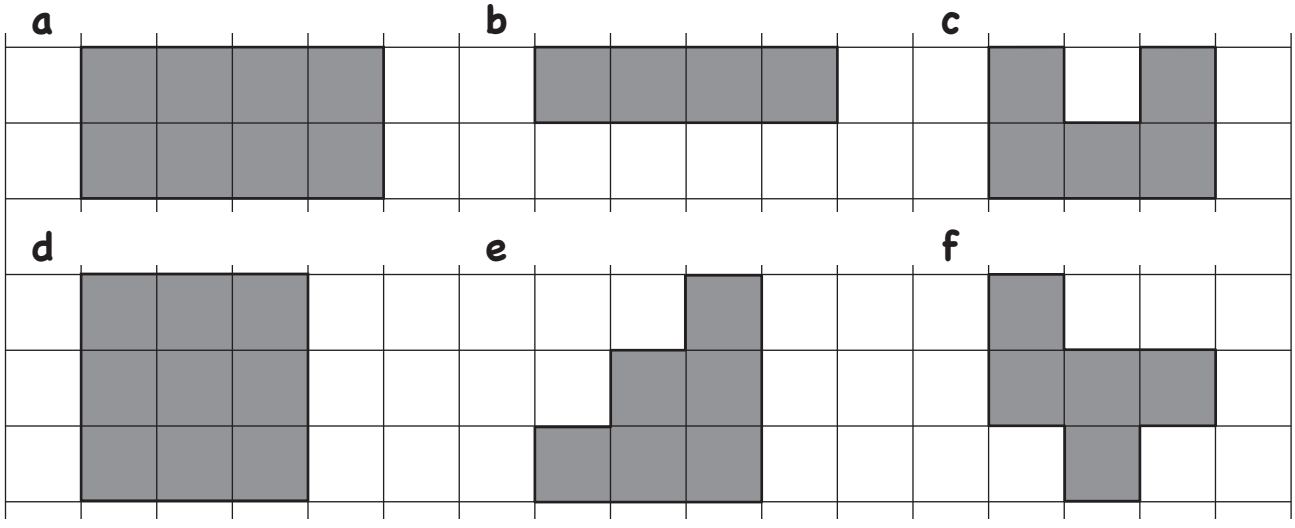
1.3 Measuring perimeters

The **perimeter** of a shape is the distance around its outside edge.

It is measured in **linear units** – millimetres, centimetres, metres or kilometres.

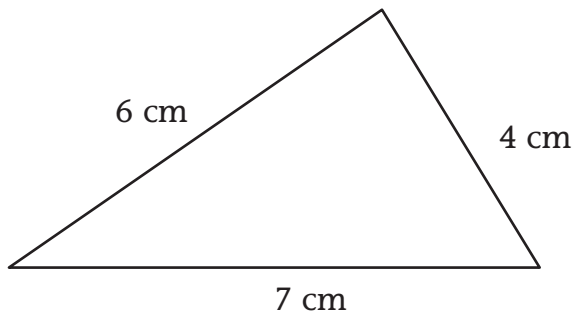


1 These shapes are drawn on a one centimetre grid.
Write the perimeter of each shape, in centimetres.



To find the **perimeter** of any shape, **add** the lengths of the sides.

EXAMPLE

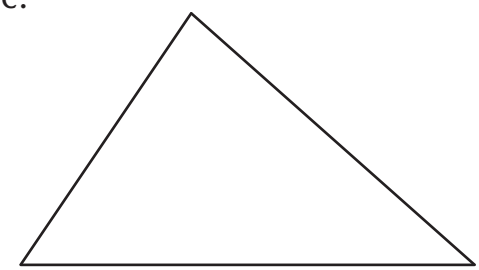


The perimeter of this triangle is:
7 cm + 6 cm + 4 cm
= 17 cm

2 Measure each side of these shapes to the nearest **centimetre**.
Then find the perimeter of each shape.

a Perimeter = _____ + _____ + _____

= _____



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1.3 Measuring perimeters

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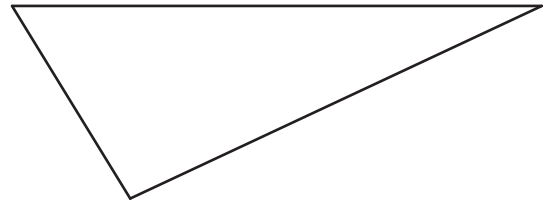
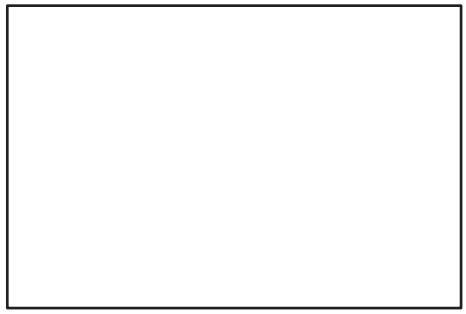
b Perimeter = _____ + _____ + _____

= _____



c

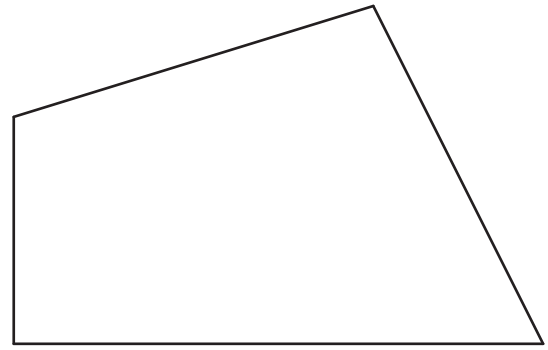


Perimeter = _____ + _____ + _____ + _____

= _____

d Perimeter = _____ + _____ + _____ + _____

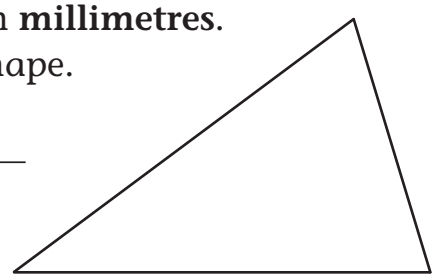
= _____



3 Measure the sides of each shape in **millimetres**. Then find the perimeter of each shape.

a Perimeter = _____ + _____ + _____

= _____



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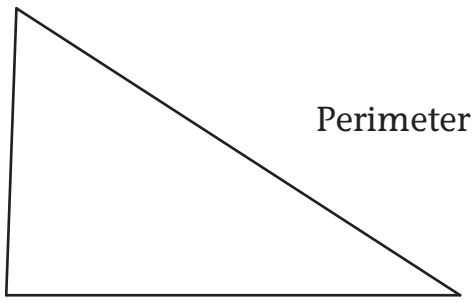


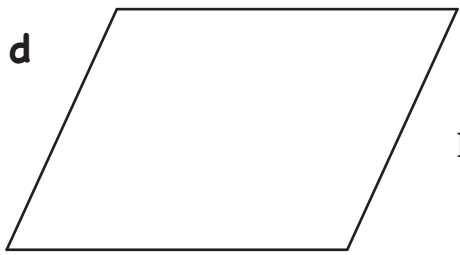
Perimeter = _____ + _____ + _____

= _____

c Perimeter = _____ + _____ + _____ + _____

= _____

d



Perimeter = _____ + _____ + _____ + _____

= _____

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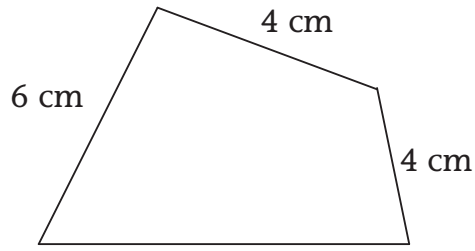
1.4 Calculating perimeters A

Terri wanted to calculate the perimeter of a piece of cardboard.
She added the lengths of the sides.

To find the *perimeter* of any shape, *add* the lengths of the sides.

EXAMPLE

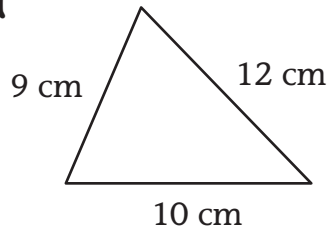


Calculate the perimeter of this shape.

The perimeter of this shape is:
7 + 6 + 4 + 4 = 21 cm

1 Calculate the perimeter of each shape.
These diagrams are ***not*** drawn to scale, so do not use your ruler.

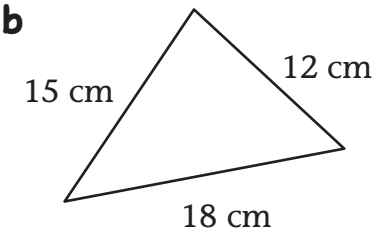
a



P = _____ + _____ + _____

= _____

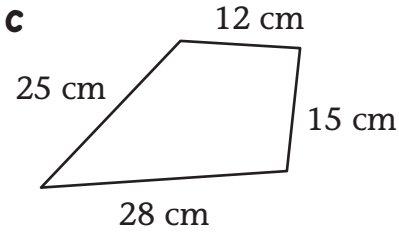
b



P = _____

= _____

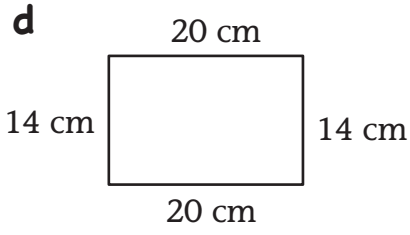
c



P = _____

= _____

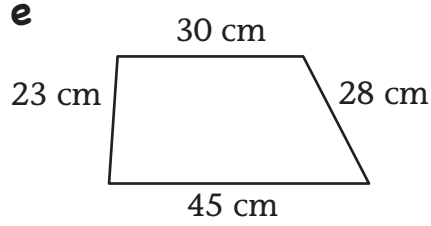
d



P = _____

= _____

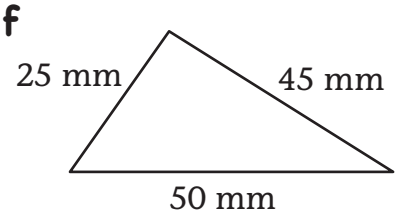
e



P = _____

= _____

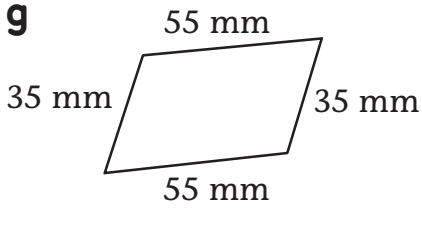
f



P = _____

= _____

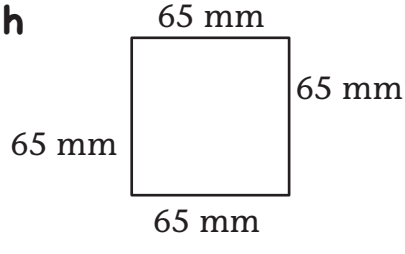
g



P = _____

= _____

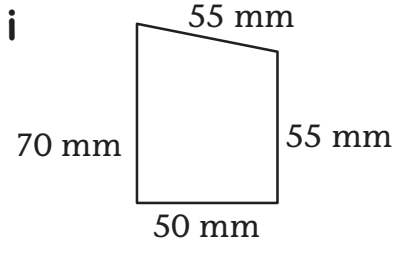
h



P = _____

= _____

i



P = _____

= _____

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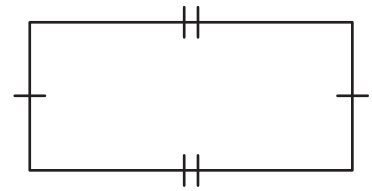
1.4 Calculating perimeters A

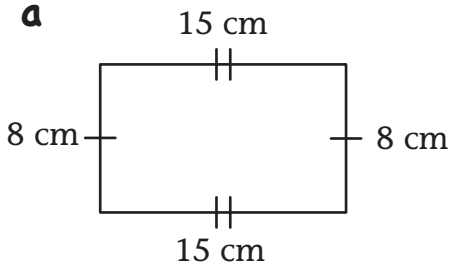
CONTINUED

On rectangles and squares, the **same marking** is used to indicate sides are the **same length**.



2 Calculate the perimeter of each rectangle and square.
The diagrams are ***not*** drawn to scale.

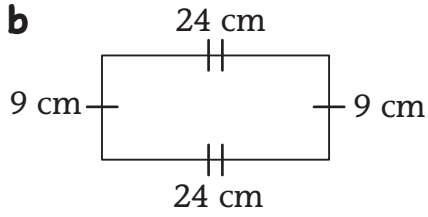
a



P = *15 + 8 + 15 + 8*

= _____

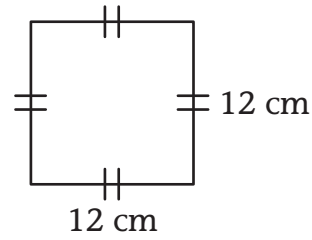
b



P = _____

= _____

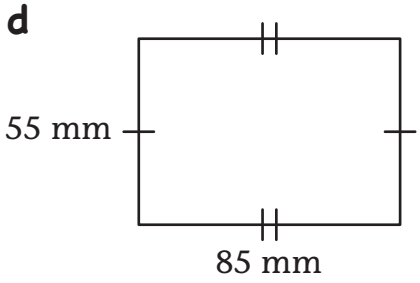
c



P = _____

= _____

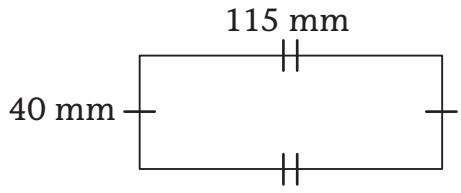
d



P = _____

= _____

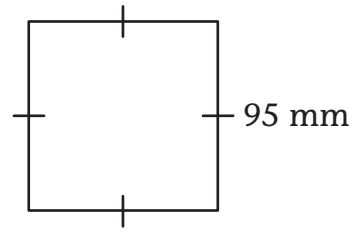
e



P = _____

= _____

f

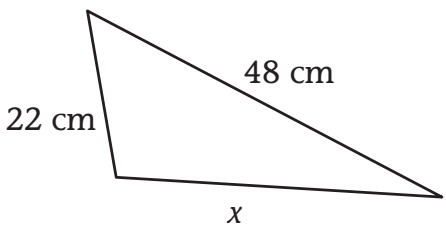


P = _____

= _____

3 On each diagram below, one length (x) is missing. You are told the perimeter of each shape. Use **subtraction** to find the missing length.

a

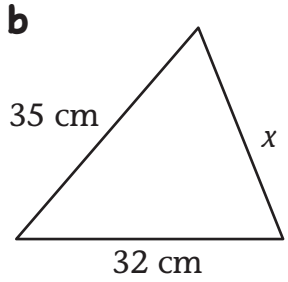


Perimeter = 105 cm

x = *105 − 48 − 22*

= _____

b

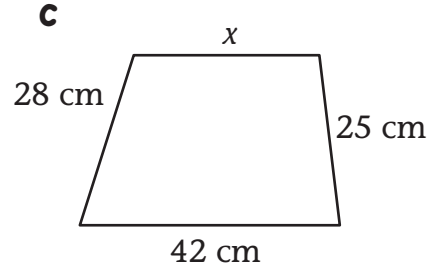


Perimeter = 95 cm

x = _____

= _____

c



Perimeter = 118 cm

x = _____

= _____

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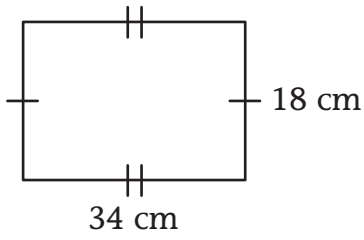
1.5 Calculating perimeters B

1 Calculate the perimeter of each rectangle and parallelogram.
Hint: opposite sides are the same length.

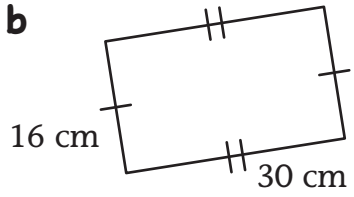
a



P = *34 + 18 + 34 + 18*

= _____

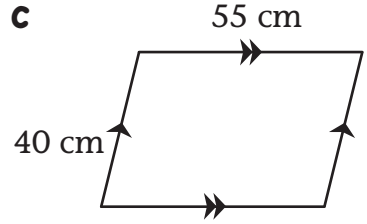
b



P = _____

= _____

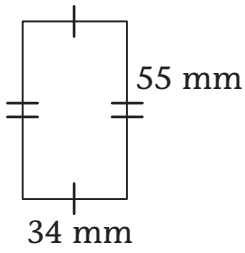
c



P = _____

= _____

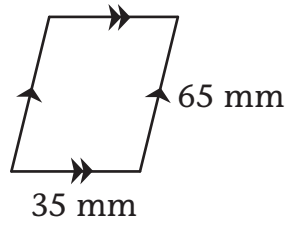
d



P = _____

= _____

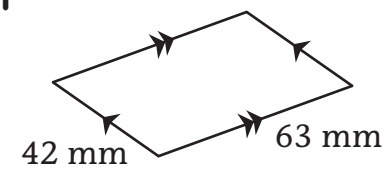
e



P = _____

= _____

f

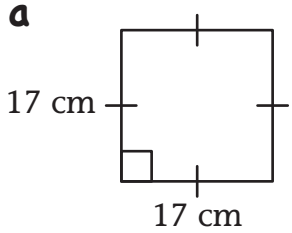


P = _____

= _____

2 Calculate the perimeter of each square and rhombus.
Hint: all squares and rhombuses have four sides of equal length.

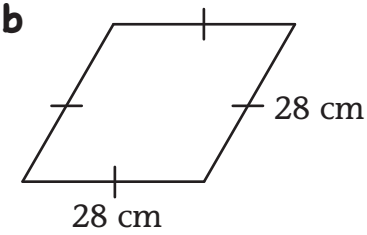
a



P = _____

= _____

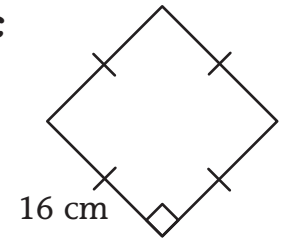
b



P = _____

= _____

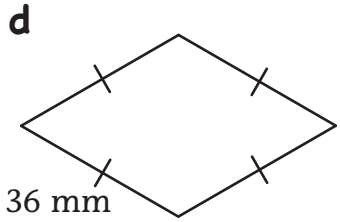
c



P = _____

= _____

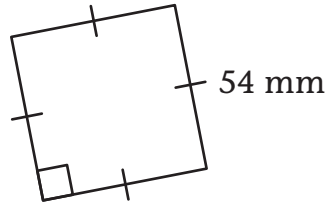
d



P = _____

= _____

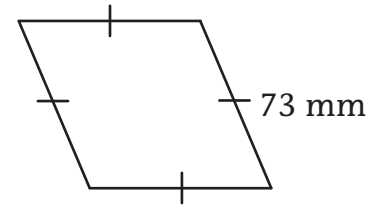
e



P = _____

= _____

f



P = _____

= _____

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Teacher's notes

OBJECTIVES

- Understand the concept of the area of a plane shape.
- Measure a variety of areas using a square centimetre grid.
- Calculate the area of rectangles and squares, using the relationship between length, breadth and area.
- Calculate the area of triangles, using the relationship between base, perpendicular height and area.
- Calculate the area of simple composite shapes.
- Select and use an appropriate formula to calculate the area of common plane shapes.

PRIOR KNOWLEDGE

Students should be able to recognise and name common plane shapes. They should be able to perform the basic operations with whole numbers, fractions and decimals using mental arithmetic or a calculator.

BACKGROUND INFORMATION

The abbreviation m^2 is read *square metre* and not 'metre squared' or 'metre square'. Also, cm^2 is read *square centimetre*. This reading applies similarly for other metric units of area.

STARTER ACTIVITY

Ask students what 'area' is in mathematics and to give examples of everyday uses of area.

Draw a simple plane shape and ask students to estimate the area of this shape. Then measure the area using a square centimetre grid overlay.

Use metre rules and large set squares to construct a square metre on a large flat surface.

ACTIVITY SHEETS

2.1 Measuring area

This activity sheet develops students' recognition of the square centimetre as a unit for measuring area. They are asked to measure a variety of areas using a square centimetre grid.

Answers

1 **a** 8 cm^2 **b** 15 cm^2 **c** 6 cm^2 **d** 16 cm^2
e 12 cm^2 **f** 9 cm^2 **g** 7 cm^2 **h** 12 cm^2
i 10 cm^2 **j** 12 cm^2 **k** 10 cm^2 **l** 8 cm^2

2 **a** 12 cm^2 **b** 12.5 cm^2 **c** 12 cm^2 **d** 6 cm^2
e 7 cm^2 **f** 12 cm^2

3 **a** 3 cm^2 **b** 21 cm^2 **c** 7.5 cm^2 **d** 12 cm^2
e 9 cm^2 **f** 5.5 cm^2

2.2 Areas of rectangles

This activity sheet develops the rule:
Area = length × width.

Students are then asked to use the given measurements to calculate the area of a number of rectangles and squares. The diagrams are not drawn to scale. The activity sheet concludes with a selection of word problems which students will need to answer in their workbooks.

Answers

1 **a** 6 cm, 1 cm, 6 cm^2 **b** 3 cm, 3 cm, 9 cm^2
c 5 cm, 3 cm, 15 cm^2 **d** 6 cm, 2 cm, 12 cm^2
e 4 cm, 3 cm, 12 cm^2

3 **a** 150 cm^2 **b** 126 cm^2 **c** 100 cm^2
d 70 cm^2 **e** 625 cm^2 **f** 540 cm^2
g 1 000 mm^2 **h** 1 300 mm^2 **i** 1 870 mm^2
j 12.96 m^2 **k** 11.76 m^2 **l** 17.82 m^2

4 405 cm^2

5 2 560 m^2

6 51.52 m^2

7 129.6 m^2

2.3 Area of triangles

This activity sheet develops the rule:
Area = $\frac{1}{2}$ × *base* × *width*.

Students are asked to use the given measurements to calculate the area of a number of triangles. These diagrams are not drawn to scale. The activity sheet concludes with a selection of word problems which students will need to answer in their workbooks.

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Teacher's notes

CONTINUED

Answers

1 **a** 16 cm^2, 8 cm^2 **b** 8 cm^2, 4 cm^2
c 12 cm^2, 6 cm^2 **d** 20 cm^2, 10 cm^2

2 The area of each triangle is **half** the area of the matching rectangle.

3 **a** 180 cm^2 **b** 180 cm^2 **c** 144 cm^2
d 640 mm^2 **e** 900 mm^2 **f** 798 mm^2
g 57.6 m^2 **h** 61.2 m^2 **i** 85.88 m^2

4 6 750 mm^2

5 14.88 m^2

6 70.125 m^2

2.4 Areas of composite shapes

This activity sheet provides a number of problems involving composite shapes. Some diagrams are divided for the students. In others, students must decide how to divide the shape into its components.

Answers

1 **a** 475 m^2 **b** 405 m^2 **c** 183 m^2

2 **a** 169 m^2 **b** 156 m^2 **c** 237 m^2

3 **a** 560 m^2 **b** 396 m^2 **c** 405 m^2

4 218 m^2

5 2.36 m^2

6 **a** 1 350 000 m^2 **b** 135 hectares

2.5 Using area formulae

In this activity sheet, students use the given formula to calculate the area of a parallelogram, trapezium, rhombus or kite. Then students must select the appropriate formula to calculate the area of a variety of plane shapes. Students will need to answer these questions in their workbooks.

Answers

1 **a** 400 cm^2 **b** 392 cm^2 **c** 468 cm^2

2 **a** 300 cm^2 **b** 378 cm^2 **c** 336 cm^2

3 **a** 176 cm^2 **b** 243 cm^2 **c** 420 cm^2

4 **a** 532 cm^2 **b** 374 cm^2 **c** 576 cm^2
d 10 625 mm^2 **e** 6 555 mm^2 **f** 7 520 mm^2
g 11.18 m^2 **h** 10.8 m^2 **i** 10.88 m^2

EXTENSION IDEAS

Provide opportunities for students to apply their measurement skills to find areas around the school, for example the area of the classroom or a basketball court.

Ask students to investigate the areas of rectangles that have the same perimeter. For example, draw two rectangles labelled with length = 6 cm, width = 4 cm and length = 7 cm, width = 3 cm. These rectangles have the same perimeter (20 cm). Ask students to calculate the areas. Then ask the question: *If I change the dimensions of a rectangle but keep the same perimeter, will the area change?*

Alternatively, ask students to compare rectangles with the same area. For example, draw two rectangles labelled with length = 10 cm, width = 4 cm and length = 8 cm, width = 5 cm. These rectangles have the same area (40 cm^2). Ask: *Do they have the same perimeter?*

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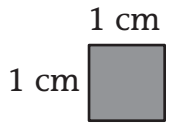
2.1 Measuring area

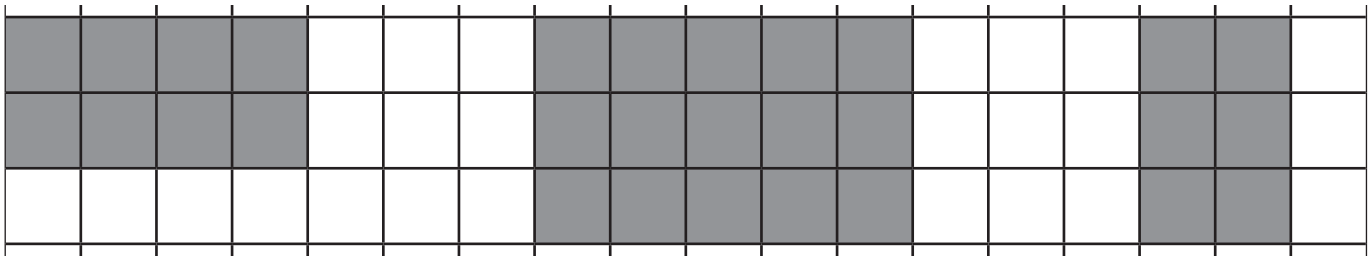
The **area** of a plane shape is the amount of the page or surface that it covers.



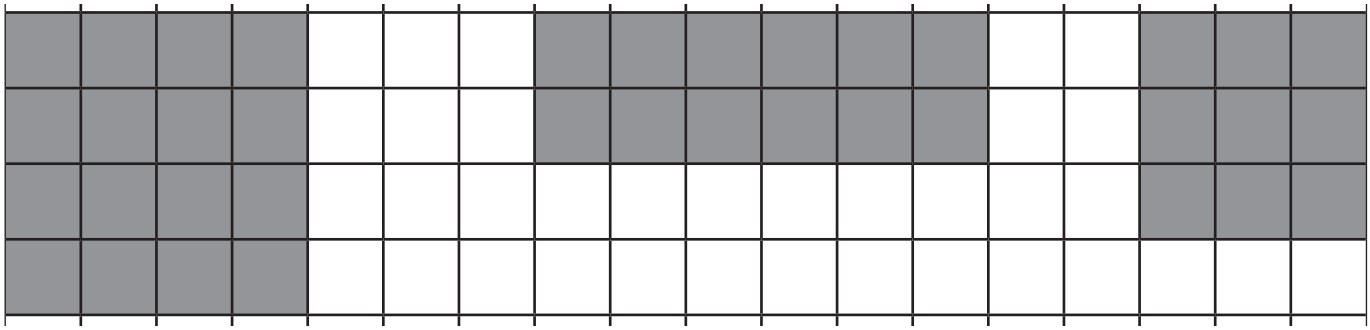
The square centimetre is a common unit to measure area.
It measures 1 cm × 1 cm. A short way of writing this unit is 1 cm^2.
We can calculate the area of a shape by counting squares.

1 The following shapes are drawn on a one centimetre grid.
Each square has an area of 1 cm^2.
What is the area of each shape? Count the number of 1 cm^2 squares.

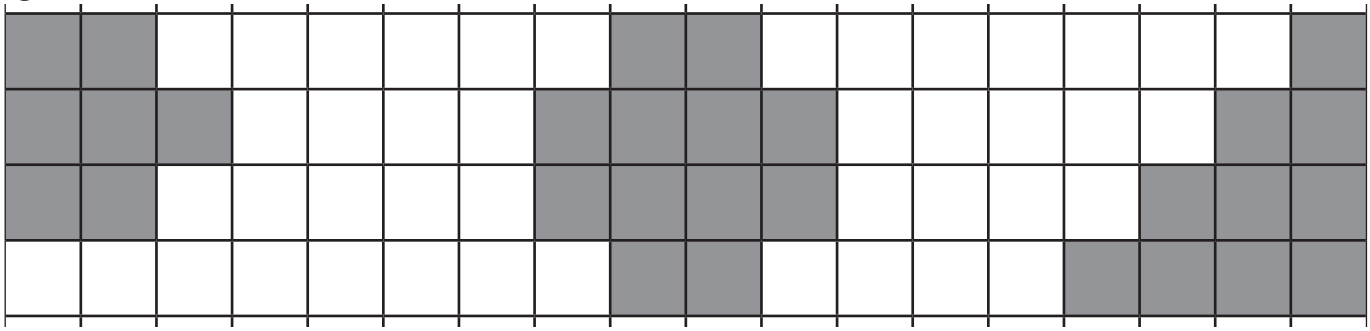
a Area = _____ **b** Area = _____ **c** Area = _____



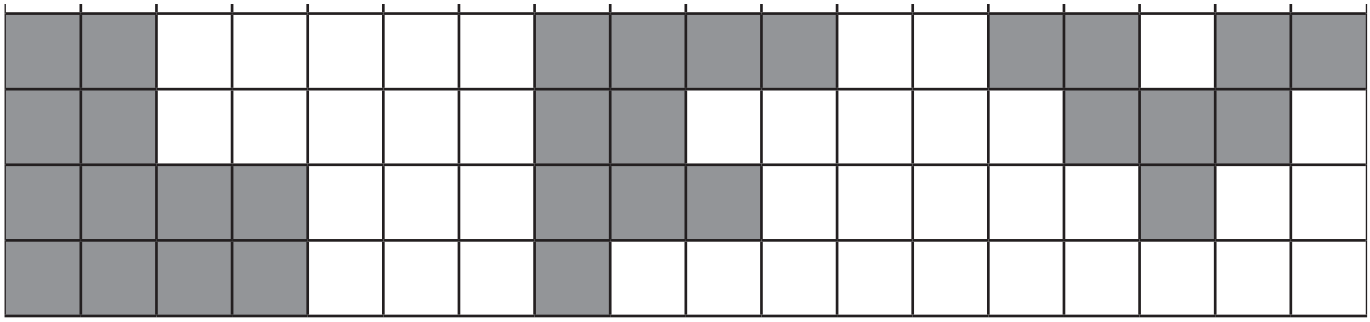
d Area = _____ **e** Area = _____ **f** Area = _____



g Area = _____ **h** Area = _____ **i** Area = _____



j Area = _____ **k** Area = _____ **l** Area = _____



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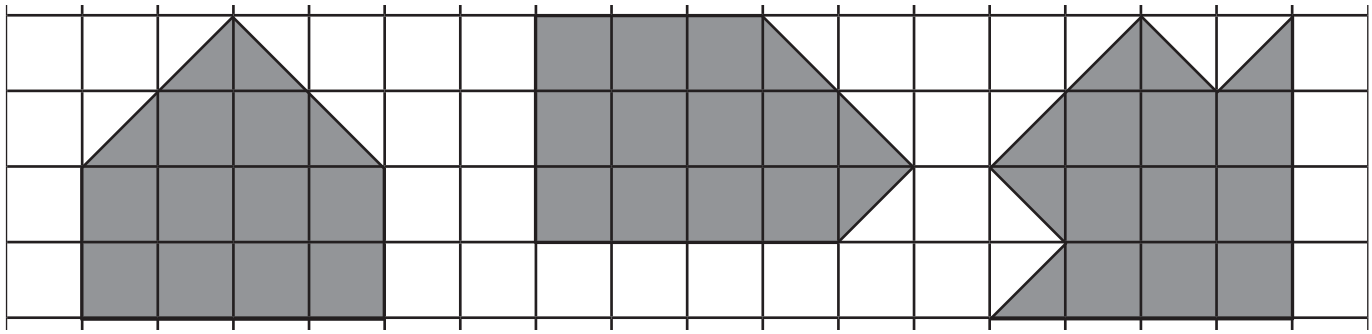
2.1 Measuring area

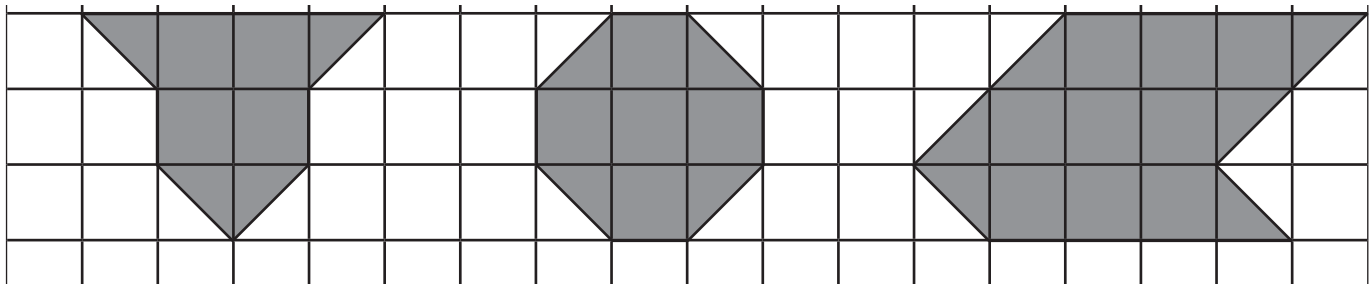
CONTINUED

2 These shapes have pieces which cover half of a square centimetre. What is the area of each shape? Count carefully.
Hint: check off the squares as you count.

a Area = _____ **b** Area = _____ **c** Area = _____

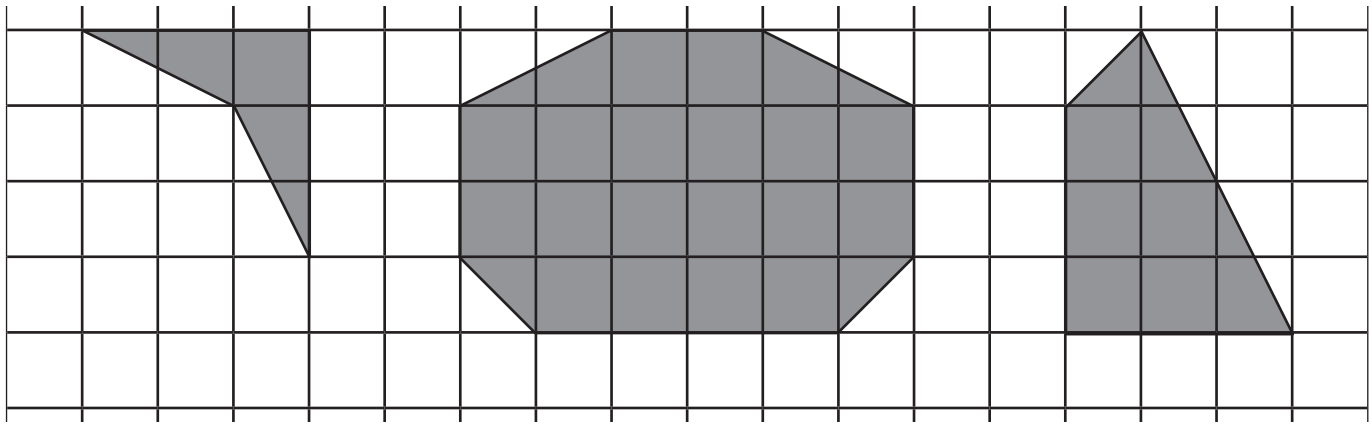


d Area = _____ **e** Area = _____ **f** Area = _____

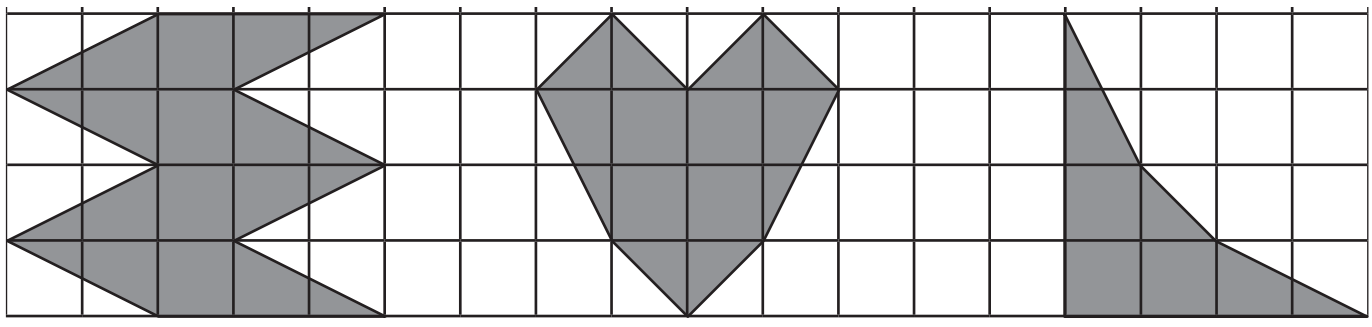


3 Find the area of each shape. Try to match pieces which make a whole square centimetre. Then count the number of squares to find the area.

a Area = _____ **b** Area = _____ **c** Area = _____



d Area = _____ **e** Area = _____ **f** Area = _____



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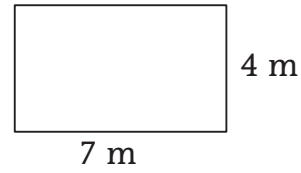
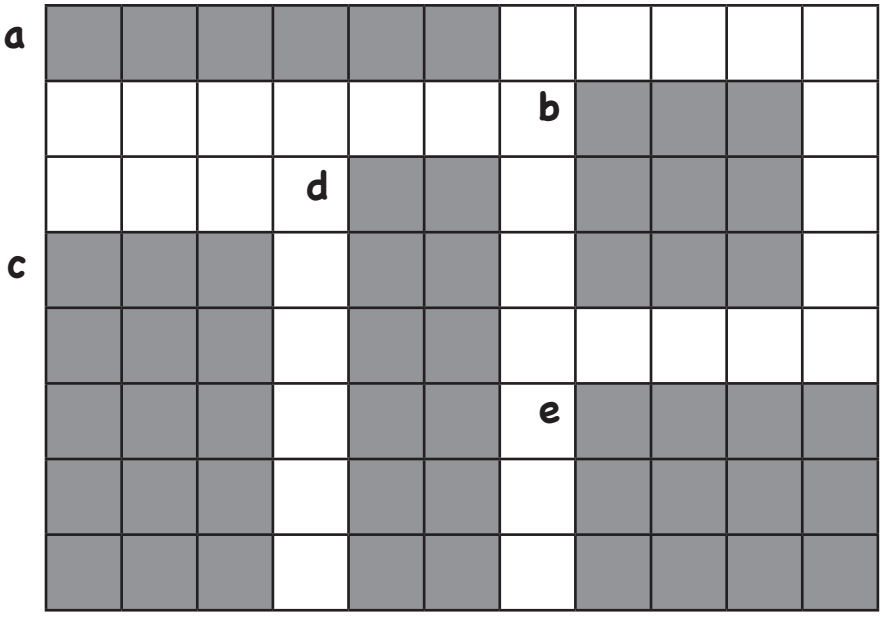
2.2 Areas of rectangles

Kym wants to buy new carpet for the living room.
The room is a rectangle measuring 7 metres by 4 metres.
Kym needs to calculate the area of this room.



1 We can calculate the area of a rectangle by counting squares.
The following rectangles are drawn on a one centimetre grid.
Write the length, width and area of each rectangle in the table provided.



	Length	Width	Area
a			
b			
c			
d			
e			

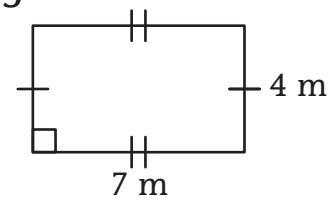
A quicker way to find the area of a rectangle is to use the rule:

Area = length × width

EXAMPLE

Find the area of this rectangle.

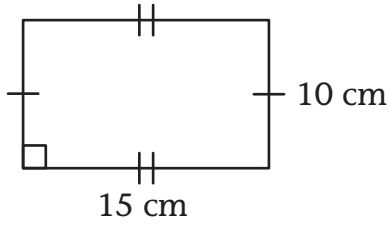
Area = length × width
= 7 × 4
= 28 cm^2



2 Use the rule ***Area* = *length* × *width*** to check your answers in question 1.

3 Use the rule ***Area* = *length* × *width*** to calculate the area of each rectangle.

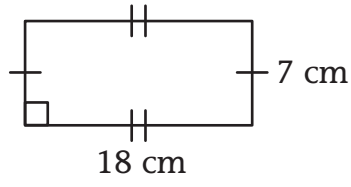
a



Area = _____ × _____

= _____

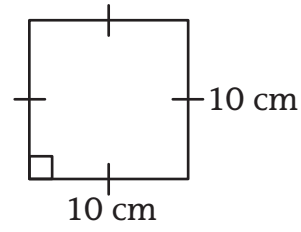
b



Area = _____

= _____

c



Area = _____

= _____

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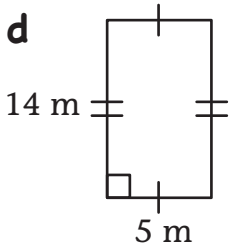
2.2 Areas of rectangles

CONTINUED

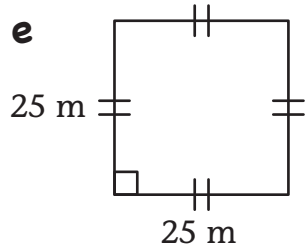
d



Area = _____

= _____

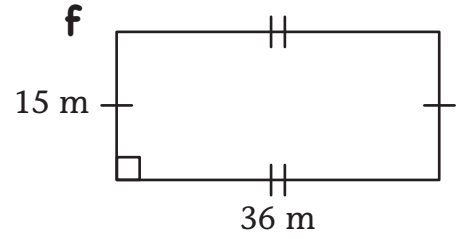
e



Area = _____

= _____

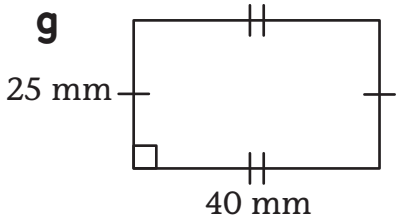
f



Area = _____

= _____

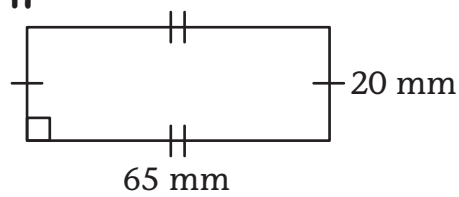
g



Area = _____

= _____

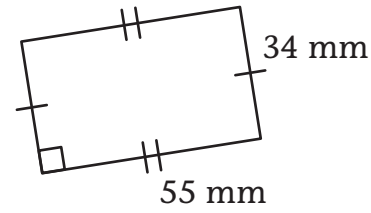
h



Area = _____

= _____

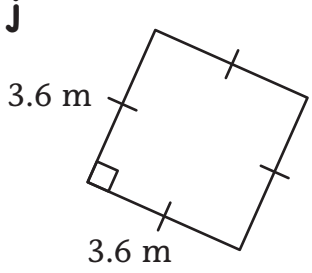
i



Area = _____

= _____

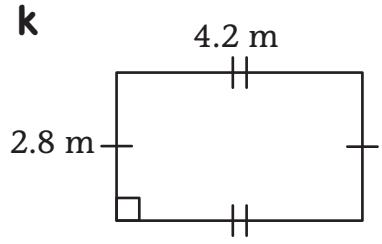
j



Area = _____

= _____

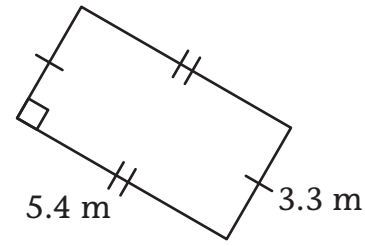
k



Area = _____

= _____

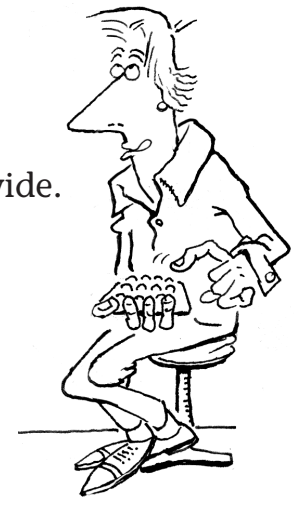
l



Area = _____

= _____

Answer the following questions in your workbook.



4 A rectangle is 27 cm long and 15 cm wide. Calculate its area.

5 A rectangular playing field is 80 metres long and 32 metres wide. Calculate its area.

6 A lawn is the shape of a rectangle.
It measures 9.2 metres by 5.6 metres.
What is its area?

7 A rectangular fence is 72 metres long and 1.8 metres high.
What is the area of one side of this fence?

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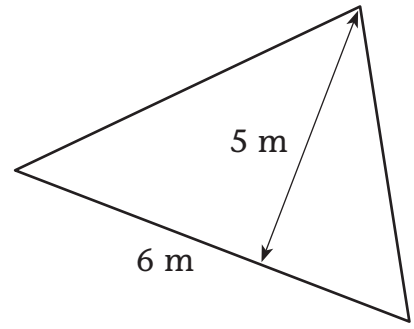
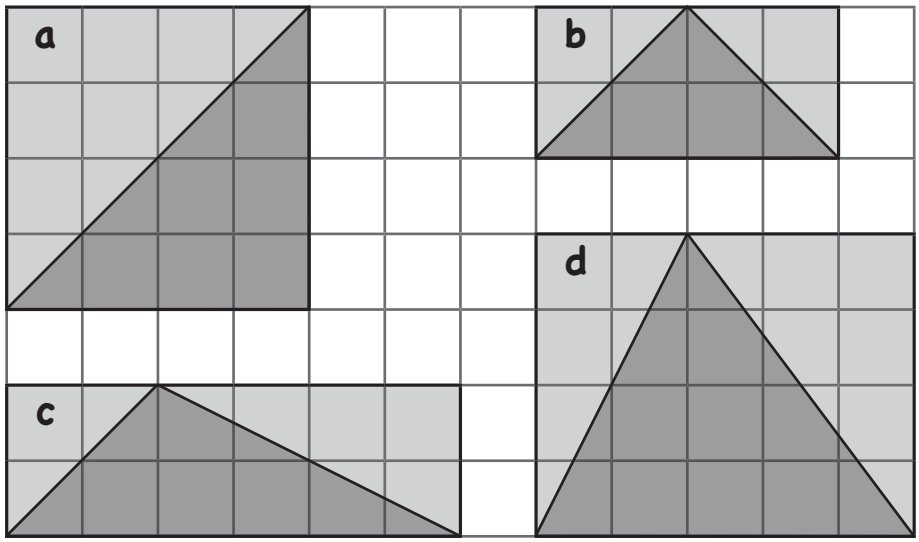
2.3 Areas of triangles

A piece of shade cloth is the shape of a triangle.
Darren wants to calculate its area.



1 In these diagrams, a triangle has been drawn inside each rectangle. The rectangles are drawn on a one centimetre grid. Find the area of each rectangle and the area of each triangle.
Write your answers in the table provided.

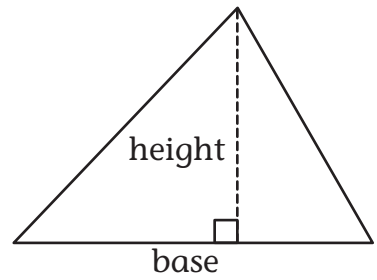


	Area of rectangle	Area of triangle
a		
b		
c		
d		

2 Compare the areas of each pair of triangles and rectangles in the table.
Complete this sentence:
The area of each triangle is _____ *the area of the matching rectangle.*

Therefore to calculate the area of a triangle,
we need the length of the base and the height.

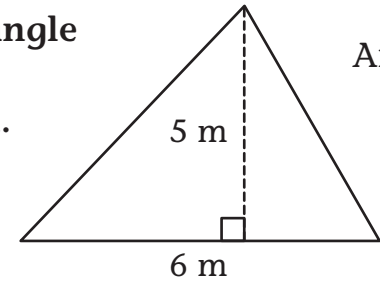
Area = $\frac{1}{2}$ × **base** × **height**



Any side of a triangle can be chosen as the base.
The height must be at **right angles** to the chosen base.

EXAMPLE

The base of a triangle is 6 m long.
The height is 5 m.
What is the area of this triangle?



Area = $\frac{1}{2}$ × base × height

= $\frac{1}{2}$ × 6 × 5

= 15 m^2

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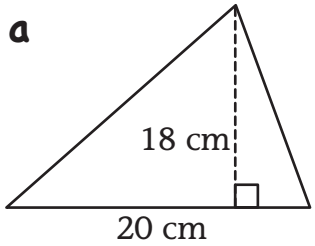
2.3 Areas of triangles

CONTINUED

3 Use the rule ***Area*** $= \frac{1}{2} \times$ ***base*** $\times$ ***height*** to calculate the area of each triangle.

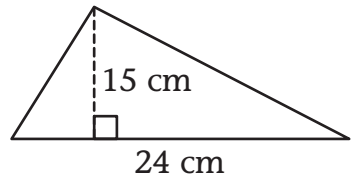
a



Area = _____

= _____

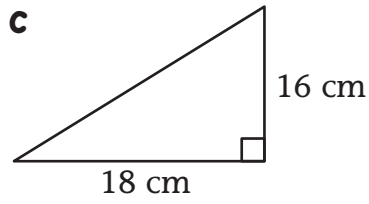
b



Area = _____

= _____

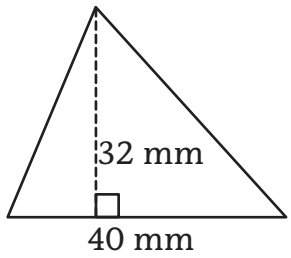
c



Area = _____

= _____

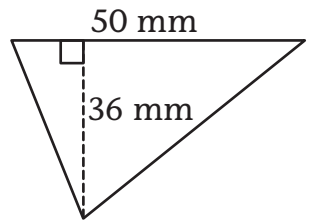
d



Area = _____

= _____

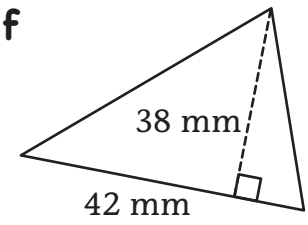
e



Area = _____

= _____

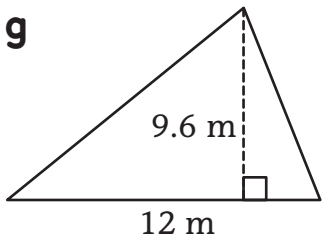
f



Area = _____

= _____

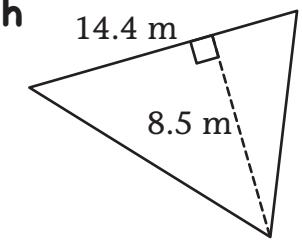
g



Area = _____

= _____

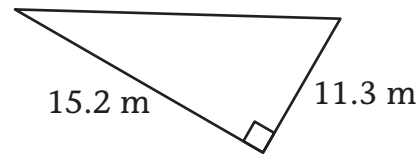
h



Area = _____

= _____

i



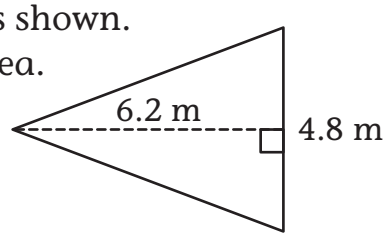
Area = _____

= _____

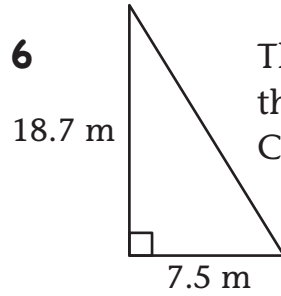
Answer the following questions in your workbook.

4 The base of a triangle measures 150 mm. Its height is 90 mm. Calculate the area of this triangle.

5 A piece of shade cloth is the shape of a triangle, as shown. Calculate its area.



6



The diagram shows the sail of a yacht. Calculate its area.

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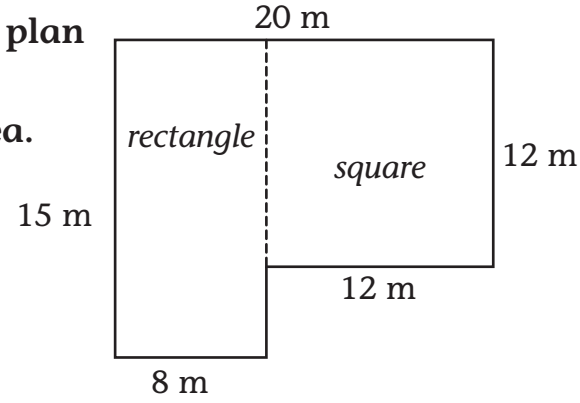
2.4 Areas of composite shapes

A **composite** shape is made up from two or more simpler shapes.

EXAMPLE

Here is the floor plan of a house.

Calculate its area.



The dotted line shows how the figure is made from two shapes.

Area = rectangle + square

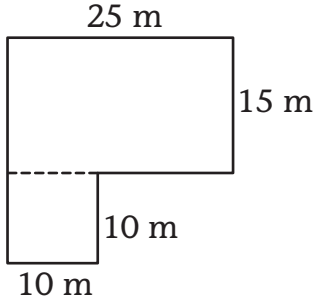
$= 15 \times 8 + 12 \times 12$

$= 120 + 144$

$= 264\ m^2$

1 Calculate the area of each composite figure.
Dotted dividing lines are shown. They divide the shapes into rectangles.

a

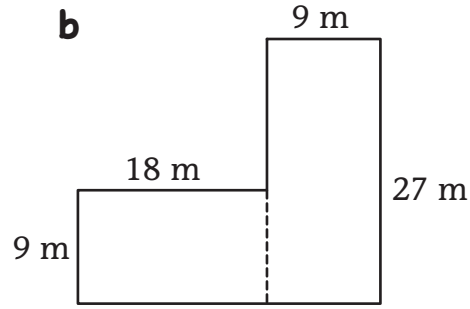


Area = _____

= _____

= _____

b

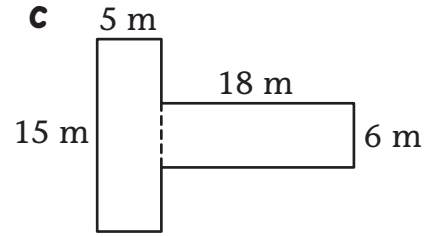


Area = _____

= _____

= _____

c



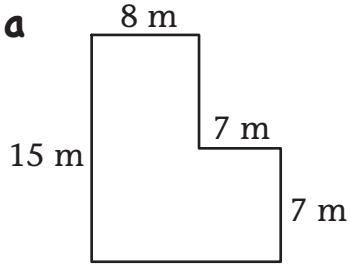
Area = _____

= _____

= _____

2 Draw lines to divide each shape into rectangles.
Then find the total area of each shape.

a

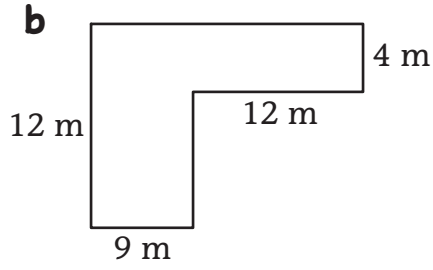


Area = _____

= _____

= _____

b

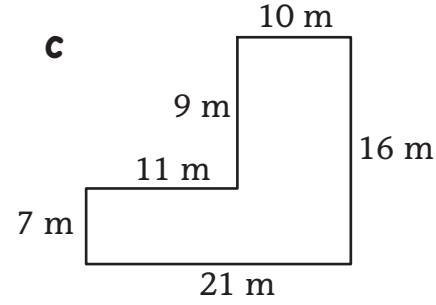


Area = _____

= _____

= _____

c



Area = _____

= _____

= _____

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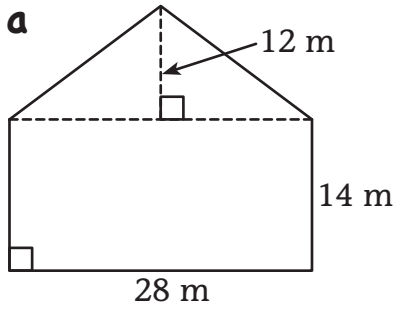
2.4 Areas of composite shapes

CONTINUED

3 The following shapes are made up of rectangles, squares and triangles. Calculate the total area of each shape.

a

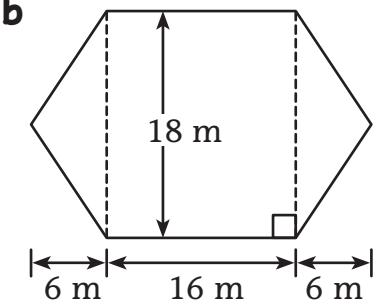


Area = _____

= _____

= _____

b

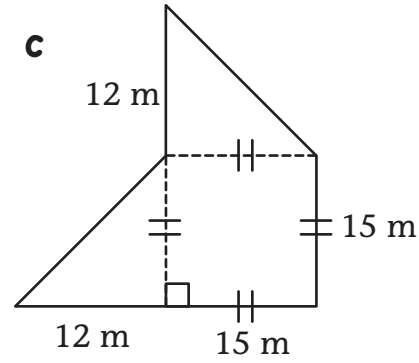


Area = _____

= _____

= _____

c



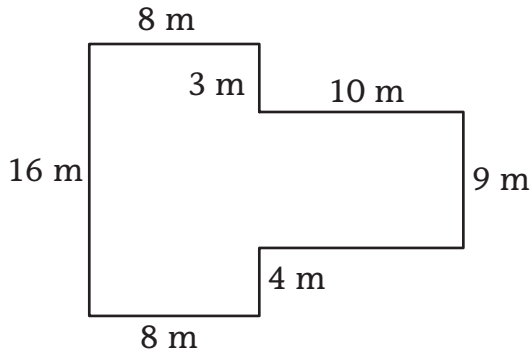
Area = _____

= _____

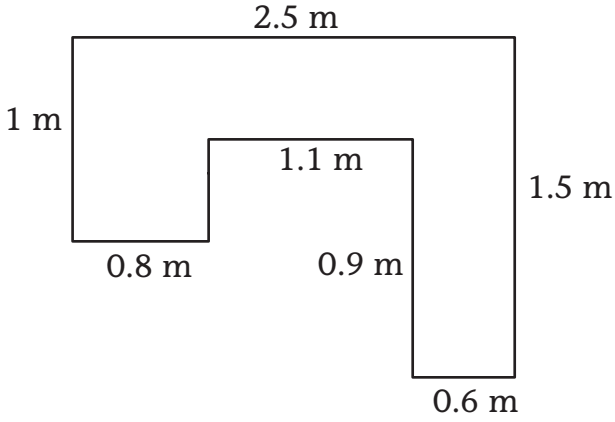
= _____

Answer the following questions in your workbook.

4 The diagram shows the plan of a house. Calculate its area.



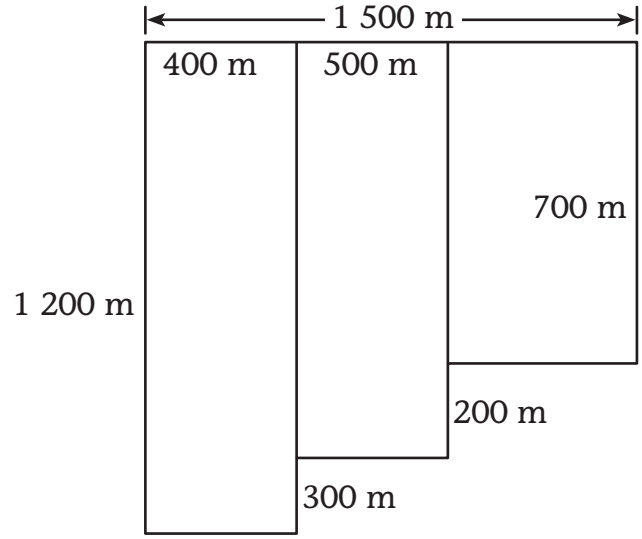
5 The top of a kitchen bench has this shape. Calculate its area.



6 A farm consists of three rectangular paddocks. What is the total area of the farm:

a in square metres?

b in hectares?
(1 hectare = 10 000 m^2)



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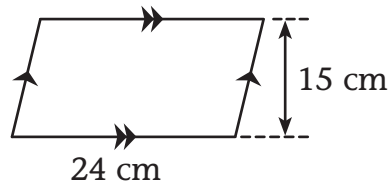
2.5 Using area formulae

There are rules (or formulae) for calculating the area of common plane shapes.

- Area of a **rectangle**
  = length × width = $l \times w$
- Area of a **parallelogram**
  = length × height = $l \times h$
- Area of a **rhombus** or **kite**
  = $\frac{1}{2}$ × (product of diagonals)
  = $\frac{1}{2} \times X \times Y$
- Area of a **triangle**
  = $\frac{1}{2}$ × base × height = $\frac{1}{2} \times b \times h$
- Area of a **trapezium**
  = $\frac{1}{2}$ × (sum of parallel sides) × height
  = $\frac{1}{2} \times (a + b) \times h$

EXAMPLE

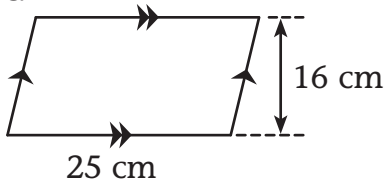
Use the formula $A = l \times h$ to calculate the area of this parallelogram.



Area = 24 × 15
= 360 cm^2

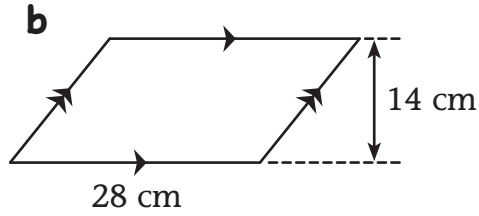
1 Use the formula $A = l \times h$ to calculate the area of each parallelogram.

a



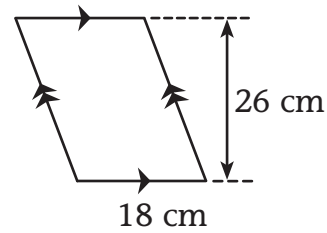
Area = _____
= _____

b



Area = _____
= _____

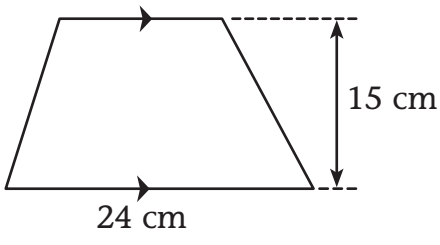
c



Area = _____
= _____

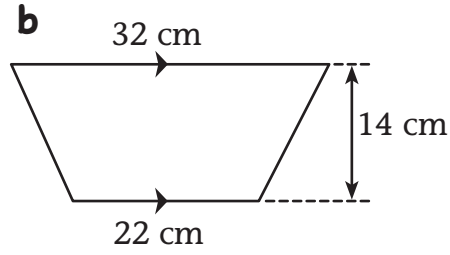
2 Use the formula $A = \frac{1}{2} \times (a + b) \times h$, to calculate the area of each trapezium.

a



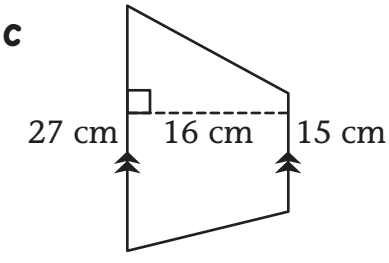
Area = _____
= _____

b



Area = _____
= _____

c



Area = _____
= _____

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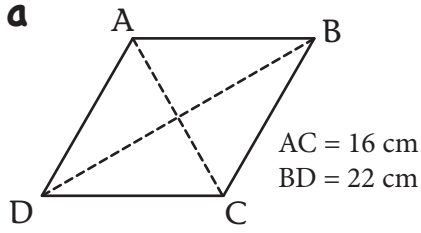
2.5 Using area formulae

CONTINUED

3 Use the formula $A = \frac{1}{2} \times X \times Y$ to calculate the area of each rhombus or kite.

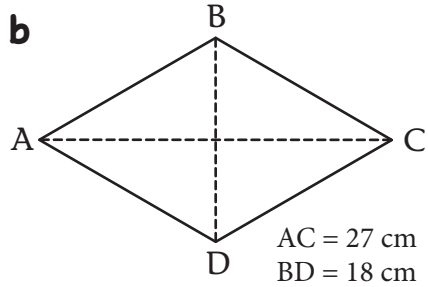
a



Area = _____

= _____

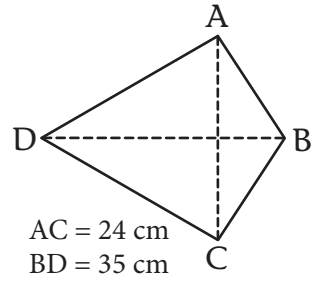
b



Area = _____

= _____

c



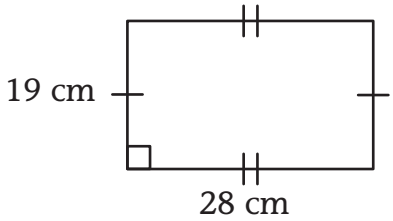
Area = _____

= _____

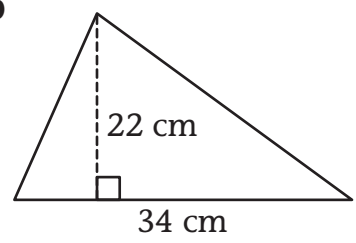
Answer the following questions in your workbook.

4 Select the correct formula and calculate the area of each plane shape.

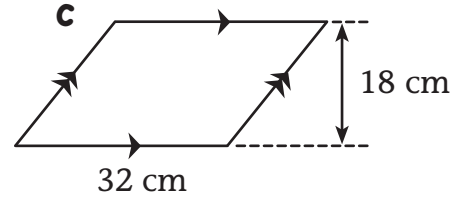
a



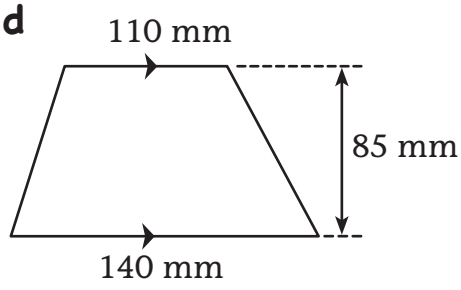
b



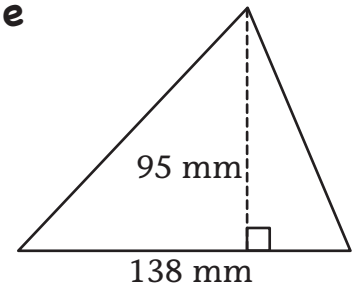
c



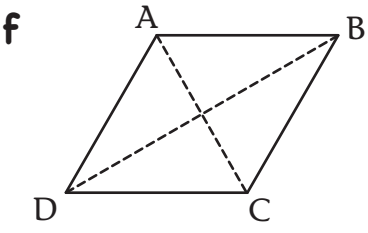
d



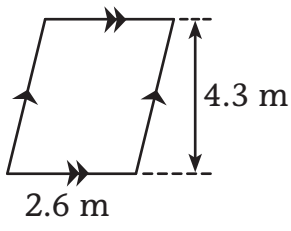
e



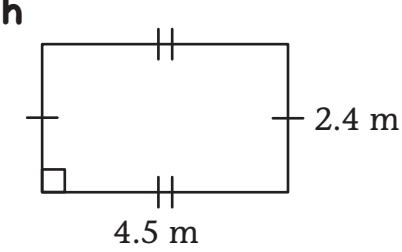
f



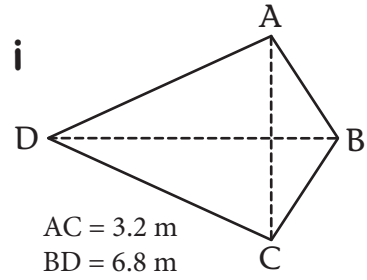
g



h



i



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Teacher's notes

OBJECTIVES

- Identify and name common three-dimensional objects, including prisms, pyramids, cylinders, cones and spheres.
- Recognise features of common solid shapes.
- Convert between different units for mass – grams, kilograms and tonnes.
- Complete calculations and solve simple problems involving mass.

PRIOR KNOWLEDGE

Students should be able to recognise and name common plane shapes which form the faces of solid shapes. They should be able to perform the basic operations with whole numbers, fractions and decimals using mental arithmetic or a calculator.

BACKGROUND

The term *solid shape* is used in geometry to refer to any three-dimensional object. The object may in fact be hollow, but is still referred to as a solid shape.

The terms *mass* and *weight* have become interchangeable in modern everyday life. However, teachers should draw students' attention to the difference.

STARTER ACTIVITY

Ask students to identify and name a selection of common solid shapes from everyday life.

Have students estimate, measure and compare the masses of a selection of solid shapes. The supermarket is a good source of regular solid shapes of varying masses and sizes.

ACTIVITY SHEETS

3.1 Solid shapes

This activity sheet provides a set of notes in the form of a cloze activity. It is important that students carefully check their answers. Encourage them to learn these key facts and terms.

Answers

1 slice, parallel, same, named, pointed, base, shape, triangles

2 vertex, face, edge, cross-section

3.2 Naming solid shapes

In this activity sheet, students name a selection of common solid shapes by matching each diagram with a name from the list provided.

Answers

1 **a** cone **b** cube **c** cylinder
d rectangular prism **e** triangular prism
f triangular pyramid **g** sphere
h square pyramid **i** hexagonal prism
j square prism **k** trapezoidal prism
l pentagonal pyramid

3.3 Faces, vertices and edges

In this activity sheet, students are asked to identify and count the faces, edges and vertices of common solid shapes. Diagrams are not provided. It can be used independently or following the previous activity sheet **3.2 Naming solid shapes**. Some students might find it useful to refer to a set of solid shapes.

Answers

1

Solid Shape	Shapes of the faces
Cube	Squares (6)
Rectangular prism	Rectangles (6)
Triangular prism	Triangles (2), Rectangles (3)
Square prism	Squares (2), Rectangles (4)
Triangular pyramid	Triangles (4)
Square pyramid	Square (1), Triangles (4)

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Teacher's notes

CONTINUED

2

Solid Shape	Faces	Vertices	Edges
Cube	6	8	12
Rectangular prism	6	8	12
Triangular prism	5	6	9
Trapezoidal prism	6	8	12
Hexagonal prism	8	12	18
Triangular pyramid	4	4	6
Square pyramid	5	5	8

3.4 Measuring mass

This activity sheet introduces the metric units for measuring mass. Students are asked to estimate the masses of objects using a supplied list of masses. They are then asked to select the most suitable units for measuring the mass of some common objects.

Answers

1 table tennis ball, tennis ball, golf ball, cricket ball, soccer ball

2 **a** 3 kg **b** 22 kg **c** 85 kg **d** 7 kg

3 **a** 680 g **b** 12 g **c** 100 g **d** 825 g

4 **a** tonnes **b** grams **c** kilograms **d** tonnes **e** kilograms **f** grams **g** kilograms **h** tonnes **i** grams **j** kilograms

3.5 Units of mass

This activity sheet develops students' skills in converting between different units for measuring mass. Encourage them to learn the rules at the top of the first page.

Answers

1 **a** 5 000 g **b** 2 600 g **c** 1 900 g **d** 900 g **e** 3 450 g **f** 25 800 g

2 **a** 8 000 kg **b** 2 900 kg **c** 5 300 kg **d** 450 kg **e** 2 750 kg **f** 18 100 kg

3 **a** 3 kg **b** 5.7 kg **c** 8.3 kg **d** 9.25 kg **e** 0.75 kg **f** 75 kg

4 **a** 7 t **b** 5.6 t **c** 3.4 t **d** 0.5 t **e** 13.9 t **f** 56 t

5 **a** 8 t **b** 7.5 kg **c** 5 900 kg **d** 8 300 g **e** 45 t **f** 3 850 g **g** 62 kg **h** 0.85 t **i** 3 650 kg **j** 250 g **k** 0.485 kg **l** 250 g **m** 950 kg **n** 2 500 g **o** 5 750 kg

6 **a** 5.7 kg **b** 1 490 kg **c** 11 550 kg **d** 4 195 kg **e** 60.9 kg **f** 20 105 kg

7 **a** 3.5 kg **b** 2.75 t **c** 3 005 g **d** 2.49 t **e** 1.25 kg **f** 0.95 t

3.6 Calculations with mass

This activity sheet provides a selection of word problems involving mass. Encourage students to write at least one line of working to show the calculation they will use to obtain each answer. Some questions require students to convert the units in their answers.

Answers

1 230 kg **2** 8.4 kg **3** 85 g **4** 1 425 kg

5 **a** 10 625 g **b** 10.625 kg

6 5.465 kg **7** 1 150 kg **8** 1 105 g **9** 850

10 320 g **11** 6.46 t **12** 15.24 kg

EXTENSION IDEAS

Have students identify and describe solid shapes in a variety of contexts such as buildings or packaging.

Students could make models of various prisms and pyramids. Encourage them to experiment with a variety of materials such as modelling clay, cardboard and craft sticks. Discuss how useful different materials are for constructing parts of shapes such as curves, edges and vertices.

Ask students to interpret and discuss the use of grams and kilograms in statements on commercial packaging. For example, on a 500 g packet of dry pasta, there is the statement that it contains 6 servings per pack. The nutrition information is based on a serving size of 85 g. Does this packet contain six 85 g servings?

Nutrition information often uses a mix of units. Converting to a consistent unit will allow easier comparison. This may also provide an opportunity to introduce milligrams (1 000 mg = 1 gram).

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3.1 Solid shapes

1 Complete these notes using the word bank.

Word bank

• base • named • pointed • parallel
• same • shape • slice • triangles

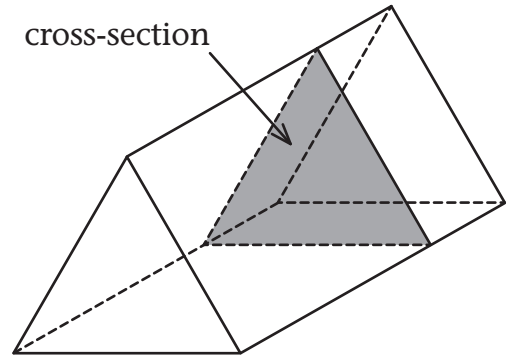
There are two main families of **solid shapes**: ***prisms*** and ***pyramids***.

A ***cross-section*** of a **prism** is made by cutting a _____ across it.

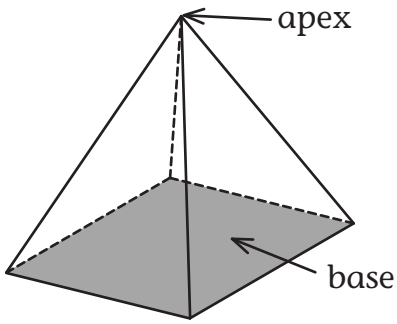
The cross-section is _____ to the end faces.

A **prism** has the same (uniform) cross-section along its length. This means a slicing machine would cut slices that are all the _____ shape and size.

A **prism** is _____ according to the shape of the cross-section.



A triangular prism



A square pyramid

A **pyramid** has a _____ top called the ***apex***.

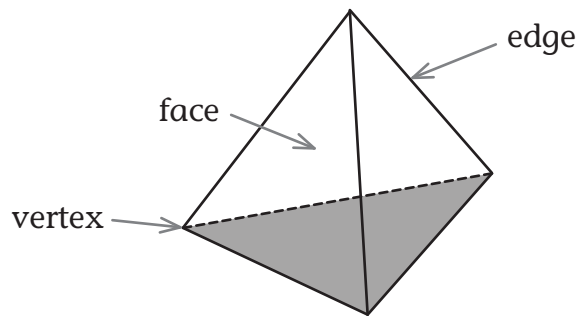
The face opposite the apex is called the _____.

Pyramids are named by the _____ of their base.

All other faces are _____.

Parts of a solid

A flat surface of a solid is called a **face**.
Two faces meet along an **edge**.
Edges meet at a **vertex**.



2 Write the correct geometrical term beside each meaning in everyday language.

Everyday language	Geometrical term
The corners of a solid shape.	
The flat surfaces of a solid shape.	
A line where two surfaces meet.	
The shape when you slice a solid shape.	

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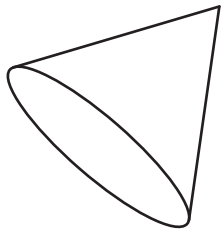
3.2 Naming solid shapes

1 Use the word bank to name each solid shape. Each name is used once only.

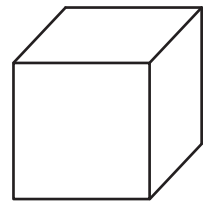
Word bank

- cone
- cube
- cylinder
- hexagonal prism
- rectangular prism
- trapezoidal prism
- triangular prism
- pentagonal pyramid
- triangular pyramid
- square prism
- square pyramid
- sphere

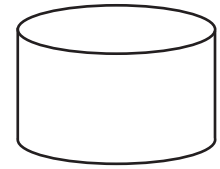
a _____



b _____

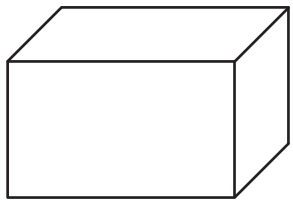


c _____



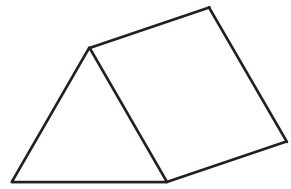
d _____

_____



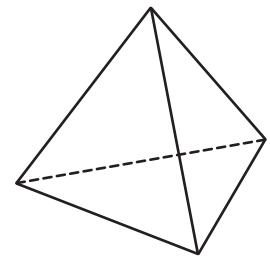
e _____

_____



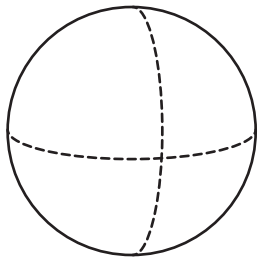
f _____

_____



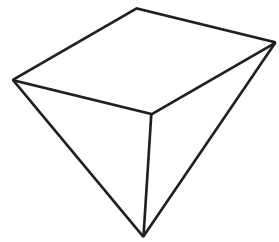
g _____

_____



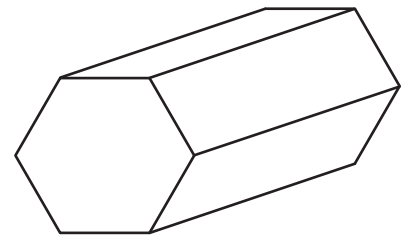
h _____

_____



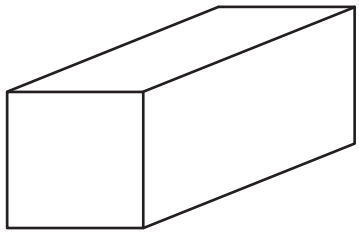
i _____

_____



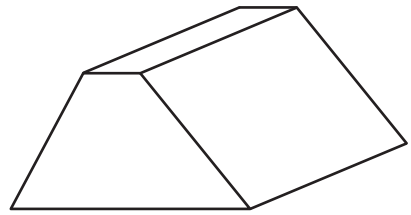
j _____

_____



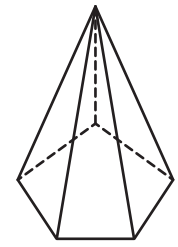
k _____

_____



l _____

_____



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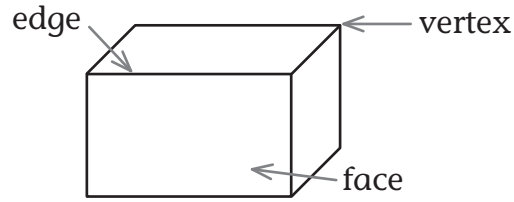
3.3 Faces, vertices and edges

A **face** is a flat surface.
An **edge** is a line where two faces meet.
A **vertex** is a corner of the solid where edges meet.

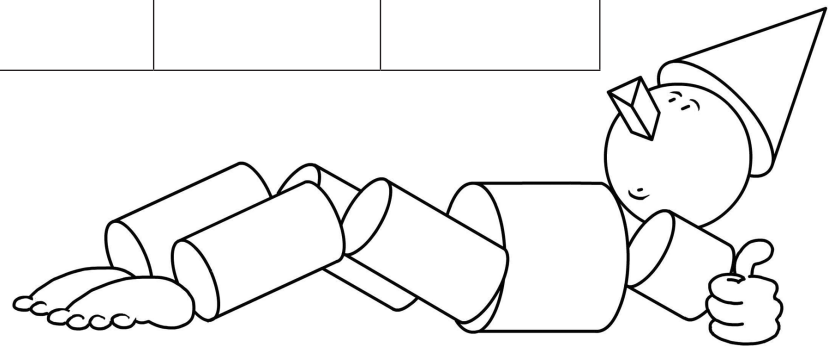


1 For each solid shape, write the two-dimensional shapes which are the faces. Write in brackets how many faces have each shape.

Solid Shape	Shapes of the faces
Cube	*squares (6)*
Rectangular prism	
Triangular prism	
Square prism	
Triangular pyramid	
Square pyramid	

2 Write the number of faces, vertices and edges on each solid shape.

Solid shape	Number of faces	Number of vertices	Number of edges
Cube	*6*	*8*	*12*
Rectangular prism			
Triangular prism			
Trapezoidal prism			
Hexagonal prism			
Triangular pyramid			
Square pyramid			



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3.4 Measuring mass

Mass measures the **amount of matter** in an object. It is the correct term for what many people call 'weight'. (*Weight* is a force exerted by gravity.)

1 Write the list of balls in order from the smallest mass to the largest mass.
golf ball, tennis ball, cricket ball, soccer ball, table tennis ball

_____

_____

In our metric system, we use the **kilogram (kg)** as the basic unit of mass.
One litre of water has a mass of exactly **1 kilogram**.

2 Write the mass which best matches each item.
Choose from: 3 kg, 7 kg, 22 kg, 85 kg.

a a new-born baby_____ **b** a 6 year-old child _____

c an adult man_____ **d** a ten-pin bowling ball_____

Other common units for measuring mass are the **gram (g)** and the **tonne (t)**.

- A paper clip has a mass of about 1 gram. 1 000 grams equal 1 kilogram.
- A small car has a mass of about 1 tonne. 1 tonne equals 1 000 kilograms.

3 Write the mass which best matches each item.
Choose from: 12 g, 100 g, 680 g, 825 g.

a a loaf of sliced bread _____ **b** a 20 cent piece_____

c an apple_____ **d** a large can of fruit_____

4 Which units – **grams**, **kilograms** or **tonnes** – do you think are most suitable for measuring the mass of:

a a family car?_____ **b** a pencil? _____

c a house brick?_____ **d** an elephant? _____

e a horse?_____

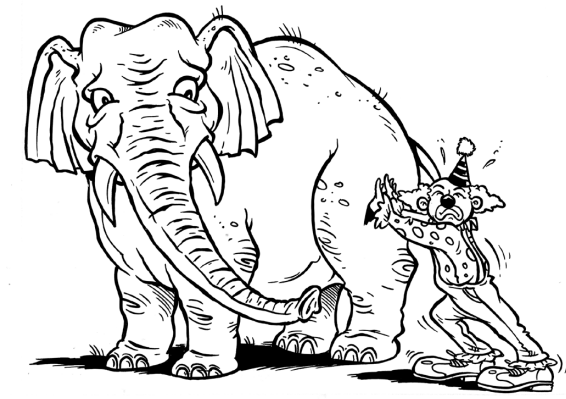
f an orange? _____

g a chicken?_____

h a bus? _____

i a pair of socks?_____

j a bag of potatoes?_____



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3.5 Units of mass

Jeremy has a mass of 50.9 kg. Dean's mass is 50 800 g.
Which boy is heavier?

To compare masses, we often need to change the units so they are the same.

1 000 grams (g) = 1 kilogram (kg)
1 000 kilograms (kg) = 1 tonne (t)

Converting to smaller units:

- To convert ***kilograms*** to ***grams***, **multiply** by 1 000.
- To convert ***tonnes*** to ***kilograms***, **multiply** by 1 000.

A quick way to **multiply** by **1 000**:
Move the decimal point 3 places to the **right**.

EXAMPLES

Convert 4 kg to grams.

4 × 1 000 = 4000 g

Convert 3.5 t to kilograms.

3.5 × 1 000 = 3500 kg

1 Convert to grams (g).

a 5 kg = _____ **b** 2.6 kg = _____ **c** 1.9 kg = _____

d 0.9 kg = _____ **e** 3.45 kg = _____ **f** 25.8 kg = _____

2 Convert to kilograms (kg).

a 8 t = _____ **b** 2.9 t = _____ **c** 5.3 t = _____

d 0.45 t = _____ **e** 2.75 t = _____ **f** 18.1 t = _____

Converting to larger units:

- To convert ***grams*** to ***kilograms***, **divide** by 1 000.
- To convert ***kilograms*** to ***tonnes***, **divide** by 1 000.

A quick way to **divide** by **1 000**:
Move the decimal point 3 places to the **left**.

EXAMPLES

Convert 6 500 g to kilograms.

6 500 ÷ 1 000 = 6.5 kg

Convert 27 000 kg to tonnes.

27 000 ÷ 1 000 = 27 t

3 Convert to kilograms (kg).

a 3 000 g = _____ **b** 5 700 g = _____ **c** 8 300 g = _____

d 9 250 g = _____ **e** 750 g = _____ **f** 75 000 g = _____

4 Convert to tonnes (t).

a 7 000 kg = _____ **b** 5 600 kg = _____ **c** 3 400 kg = _____

d 500 kg = _____ **e** 13 900 kg = _____ **f** 56 000 kg = _____

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3.5 Units of mass

CONTINUED

5 Complete the following.

a 8 000 kg = _____ t **b** 7 500 g = _____ kg **c** 5.9 t = _____ kg

d 8.3 kg = _____ g **e** 45 000 kg = _____ t **f** 3.85 kg = _____ g

g 62 000 g = _____ kg **h** 850 kg = _____ t **i** 3.65 t = _____ kg

j 0.25 kg = _____ g **k** 485 g = _____ kg **l** $\frac{1}{4}$ kg = _____ g

m 0.95 t = _____ kg **n** $2\frac{1}{2}$ kg = _____ g **o** $5\frac{3}{4}$ t = _____ kg

Sometimes we need to **compare two different masses** that are expressed in different units.

EXAMPLE

Jeremy has a mass of 50.9 kg. Dean's mass is 50 800 g. Which boy is heavier?

The first step is convert to the same units.

Convert 50.9 kg to 50 900 g ***or*** convert 50 800 g to 50.8 kg.

We can now compare the same units.

Compare 50 900 g with 50 800 g ***or*** compare 50.9 kg with 50.8 kg

Answer: Jeremy is heavier.

6 For each pair of masses, ***circle*** the **larger** mass.

a 5 600 g or 5.7 kg **b** 1 490 kg or 1.4 t

c 11.5 t or 11 550 kg **d** 4 195 g or 4.19 kg

e 60.9 kg or 60 848 g **f** 20.1 t or 20 105 kg

7 For each pair of masses, ***circle*** the **smaller** mass.

a 3 600 g or 3.5 kg

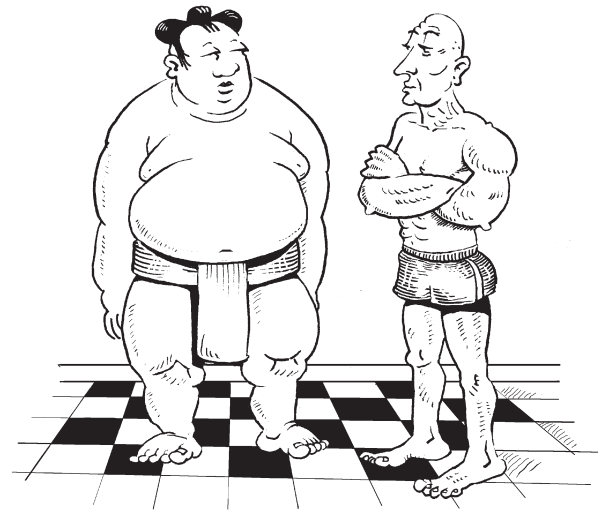
b 2.75 t or 2 752 kg

c 3.05 kg or 3 005 g

d 2 495 kg or 2.49 t

e 1.25 kg or 125 500 g

f 0.95 t or 955 kg



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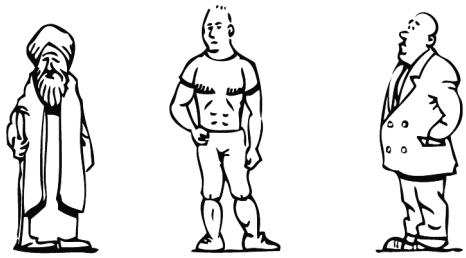
3.6 Calculations with mass



1 Three men weigh 68 kg, 79 kg and 83 kg. What is their total mass?

_____

2 Paul bought 1.2 kg of apples, 1.6 kg of oranges, 2.1 kg of beans and 3.5 kg of potatoes. What is the total mass of his shopping?

_____

3 Find the total mass of 50 nails if the mass of each nail is 1.7 grams.

_____

4 Calculate the total mass in kilograms of 15 sheets of metal. Each sheet of metal has a mass of 95 kg.

_____

5 A box of breakfast cereal has a mass of 425 g. What is the total mass of 25 boxes:

a in grams? _____

b in kilograms? _____

6 Calculate the total mass in kilograms of five parcels of mass 860 g, 435 g, 920 g, 1.5 kg and 1.75 kg.

_____

7 Charley bought 5 tonnes of garden mulch. She used 3 850 kg. What mass in kilograms remains?

_____

8 Jon's dad bought $1\frac{1}{2}$ kg of nuts. Jon and his friends ate 395 g of them. What mass is left?

_____

9 The nails in a box have a total mass of 1.53 kg. Each nail has a mass of 1.8 grams. How many nails are in the box?

_____

10 A box contains 24 mangoes. The total mass of the mangoes is 7.68 kg. What is the average mass of each mango, in grams?

_____

11 A truck is loaded with 1 700 bricks. Each brick has a mass of 3.8 kg. What is the mass of the load in tonnes?

_____

12 A cardboard box weighs 240 grams. It is packed with 10 bags of flour. Each bag weighs 1.5 kg. What is the total weight of the packed box?

_____

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Teacher's notes

OBJECTIVES

- Understand the concept of the volume of a solid shape.
- Understand the concept of the capacity of a container.
- Convert between units for capacity – millilitres, litres and kilolitres.
- Calculate the volume of a rectangular prism using the relationship between length, breadth, height and volume.
- Calculate the volume of various prisms using the formula $V = A \times h$.
- Calculate the capacity of a container.

PRIOR KNOWLEDGE

Students should be able to recognise and name common solid shapes. They should be able to perform the basic operations with whole numbers, fractions and decimals using mental arithmetic or a calculator.

BACKGROUND

Volume refers to the amount of space occupied by an object or substance. Capacity refers to the amount a container can hold. Capacity is only used in relation to containers.

The abbreviation cm^3 is read *cubic centimetre* and not 'centimetre cubed'.

STARTER ACTIVITY

Ask students to estimate and measure the volumes of various containers by filling them with cubes.

Build a cubic metre in the classroom. Brainstorm appropriate building materials and work with the class to construct it. You could discuss the fact that 1 000 000 cubic centimetres fit inside a cubic metre.

ACTIVITY SHEETS

4.1 Volume of solid shapes

This activity sheet develops students' recognition of the cubic centimetre as a unit for measuring volume. They are asked to determine the volume of a selection of solid shapes by counting cubes. Each cube has a volume of 1 cm^3.

Answers

1 a 4 cm^3 **b** 3 cm^3 **c** 4 cm^3 **d** 6 cm^3
e 8 cm^3 **f** 6 cm^3 **g** 12 cm^3 **h** 8 cm^3
i 18 cm^3

4.2 Volume of a rectangular prism A

This activity sheet develops the rule:
Volume = length × width × height.
Students start by counting the number of cubes in each layer. Then they multiply by the number of layers.

Answers

1 a 12 cm^3 **b** 36 cm^3 **c** 30 cm^3 **d** 36 cm^3
e 32 cm^3 **f** 125 cm^3

4.3 Volume of a rectangular prism B

In this activity sheet students are asked to calculate the volume of a number of rectangular prisms using the formula and the given dimensions. These diagrams are not drawn to scale. The activity sheet concludes with a selection of word problems. Students will need to answer these questions in their workbooks.

Answers

1 a 192 cm^3 **b** 1 344 cm^3 **c** 2 880 cm^3
d 1 694 cm^3 **e** 2 160 cm^3 **f** 5 760 cm

2 400 m^3 **3** 13.72 m^3 **4** 799 200 cm^3

4.4 Volume of a prism

This activity sheet develops the rule:
Volume = Area of face × height.
Students are then asked to calculate the area of a number of prisms using the given measurements. These diagrams are not drawn to scale.

Answers

1 a 300 cm^3 **b** 420 cm^3 **c** 576 cm^3
d 414 cm^3 **e** 672 cm^3 **f** 630 cm^3
g 1 300 cm^3 **h** 532 cm^3 **i** 1 728 cm^3
j 992 cm^3 **k** 540 cm^3 **l** 858 cm^3

2 a 144 cm^2, 2 592 cm^3 **b** 176 cm^2, 2 640 cm^3
c 384 cm^2, 5 376 cm^3 **d** 126 cm^2, 3 024 cm^3
e 72 cm^2, 1 656 cm^3 **f** 175 cm^2, 3 325 cm^3

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Teacher's notes

CONTINUED

4.5 Measuring capacity

This activity sheet introduces the metric units for measuring capacity. Students are asked to estimate the capacities of containers using a supplied list of capacities. They are then asked to select the most suitable units for measuring the capacity of some common objects.

Answers

1 **a** 7 L **b** 2 L **c** 180 L **d** 70 L

2 **a** 375 mL **b** 25 mL **c** 5 mL **d** 2 L
e 400 L **f** 60 kL **g** 500 mL **h** 200 mL

3 **a** litres **b** kilolitres **c** millilitres
d kilolitres **e** litres **f** litres **g** millilitres
h litres **i** litres **j** kilolitres

4.6 Units of capacity

In this activity, students' skills in converting between different units for measuring capacity are developed. Encourage them to learn the given rules.

Answers

1 **a** 4 000 mL **b** 14 000 mL **c** 1 600 mL
d 1 250 mL **e** 800 mL **f** 950 mL

2 **a** 3 000 L **b** 50 000 L **c** 4 400 L
d 12 500 L **e** 1 250 L **f** 850 L

3 **a** 3 L **b** 1.3 L **c** 2.25 L **d** 12.5 L
e 0.6 L **f** 0.75 L

4 **a** 4 kL **b** 2.5 kL **c** 6.25 kL **d** 12 kL
e 15.6 kL **f** 0.95 kL

4.7 Calculations with capacity

This activity sheet provides a selection of word problems involving capacity. Encourage students to write at least one line of working to show the calculations they will use to obtain their answer.

Answers

1 **a** 56 L **b** 6.25 L **c** 3 L **d** 7.5 L

2 1050

3 9.29

4 8

5 $7\frac{1}{2}$

6 9

7 **a** 2 litre bottle
b 6 bottles each containing 450 mL

4.8 Capacity and volume A

This activity sheet develops the link between volume and capacity.

Answers

1 **a** 400 mL **b** 320 mL **c** 1 250 mL
d 850 cm^3 **e** 95 cm^3 **f** 375 cm^3

2 **a** 7 L **b** 3.2 L **c** 0.65 L **d** 3 000 cm^3
e 8 400 cm^3 **f** 250 cm^3

3 **a** 6 000 L **b** 3 900 L **c** 700 L **d** 5 m^3
e 2.6 m^3 **f** 0.85 m^3

4 68 L

5 2 850 L

6 35 m^3

4.9 Capacity and volume B

This activity sheet provides a selection of problems involving volume and capacity. Encourage students to write at least one line of working to show the calculation they will use to obtain their answer.

Answers

1 **a** 1 200 cm^3, 1 200 mL **b** 4 000 cm^3, 4 000 mL **c** 2 160 cm^3, 2 160 mL

2 **a** 72 000 cm^3, 72 L **b** 90 000 cm^3, 90 L
c 87 500cm^3, 87.5L

3 **a** 512 m^3 **b** 512 000 L **c** 512 kL

4 **a** 3 375cm^3 **b** 3 375 mL

5 **a** 80 m^3 **b** 80 000 L

6 **a** 105 000 cm^3 **b** 105 L

7 **a** 520 000 cm^3 **b** 520 litres

EXTENSION IDEAS

Challenge students to use centimetre cubes to construct at least five different rectangular prisms that have the same volume.

Ask students to interpret information about capacity and volume on commercial packaging.

Students could solve problems relating to packing articles in a carton. For example:

A cardboard carton measures 40 cm × 30 cm × 25 cm. It contains ten 2 litre bottles of orange cordial. How much empty space is left in the carton?

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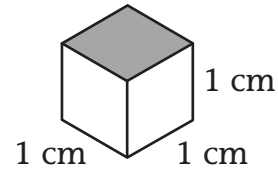
4.1 Volume of solid shapes

Volume refers to the amount of space occupied by a solid shape.

It is measured in cubic units.

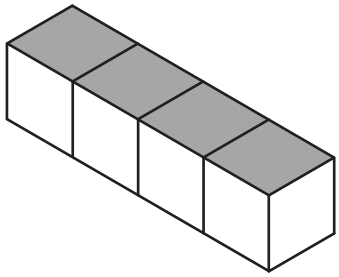
Commonly used metric units for volume are cubic millimetres (mm^3), cubic centimetres (cm^3) and cubic metres (m^3).

A cube with edges one centimetre (1 cm) long has a volume of 1 cm^3.



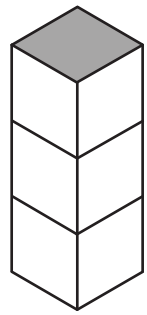
1 Count the number of cubes to find the volume of each solid shape. Each cube has a volume of 1 cm^3.

a



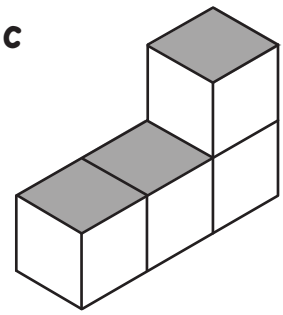
Volume = _____

b



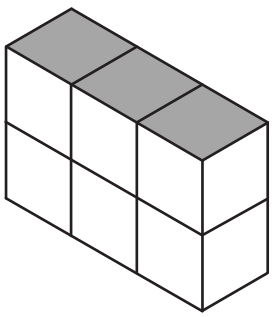
Volume = _____

c



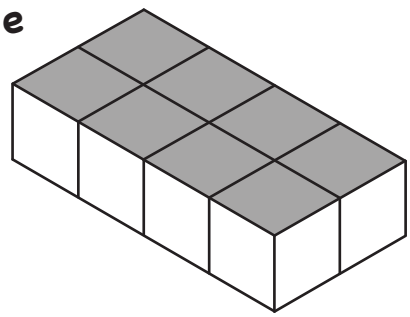
Volume = _____

d



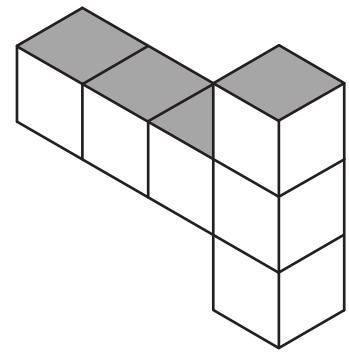
Volume = _____

e



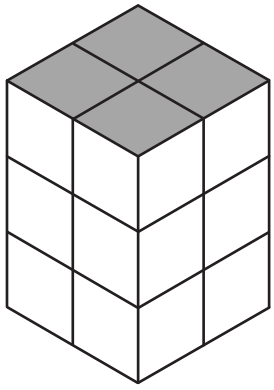
Volume = _____

f



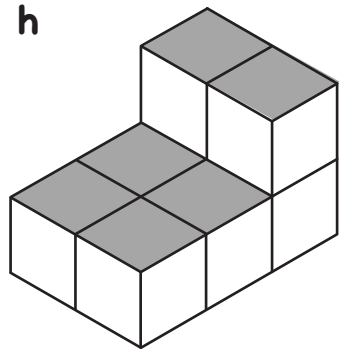
Volume = _____

g



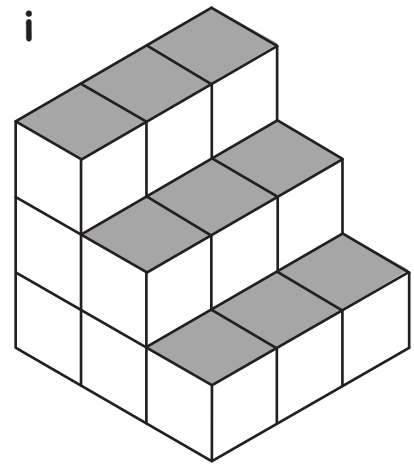
Volume = _____

h



Volume = _____

i



Volume = _____

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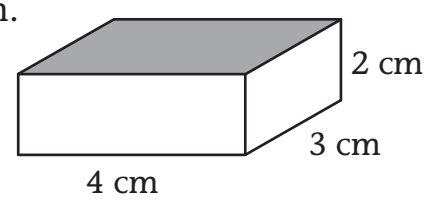
4.2 Volume of a rectangular prism A

A small box is the shape of a rectangular prism.
It is 4 cm long, 3 cm wide and 2 cm deep.
Binh wants to calculate the volume.

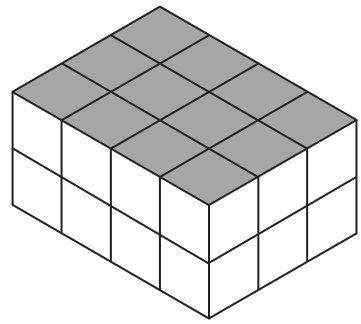


EXAMPLE

This rectangular prism is made using 1 cm cubes.

Each cube has a volume of 1 cm^3.
The prism has two layers.

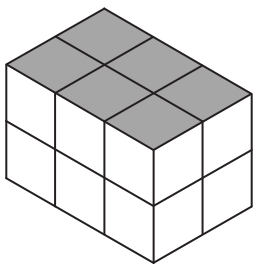
Each layer has 12 cubes
(3 rows of 4 cubes).



Volume of this prism
= 12 × 2
= 24 cm^3

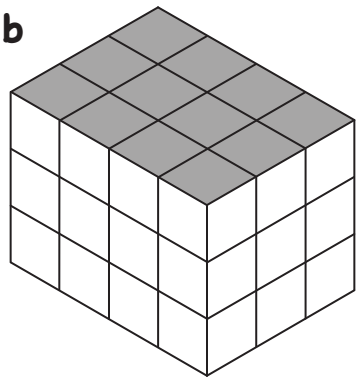
1 These rectangular prisms are made using 1 cm cubes.
Count the number of cubes in one layer. Then multiply by the number of layers. This will give the volume of the prism.

a



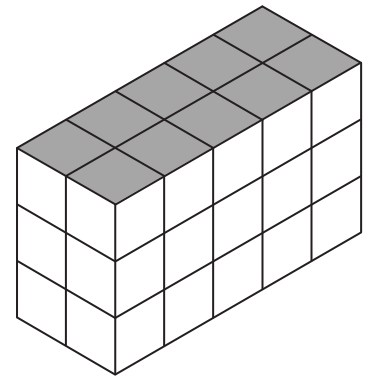
Volume = _____

b



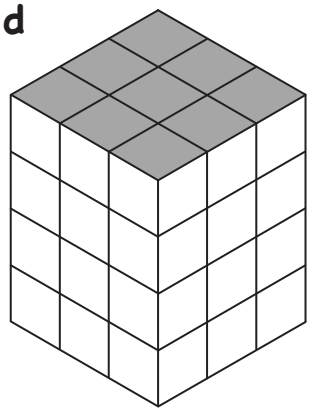
Volume = _____

c



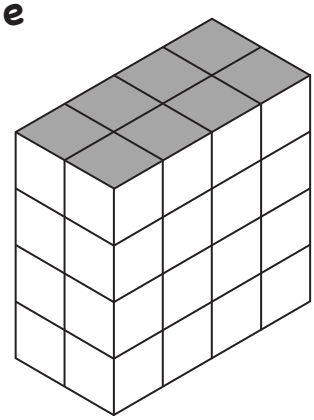
Volume = _____

d



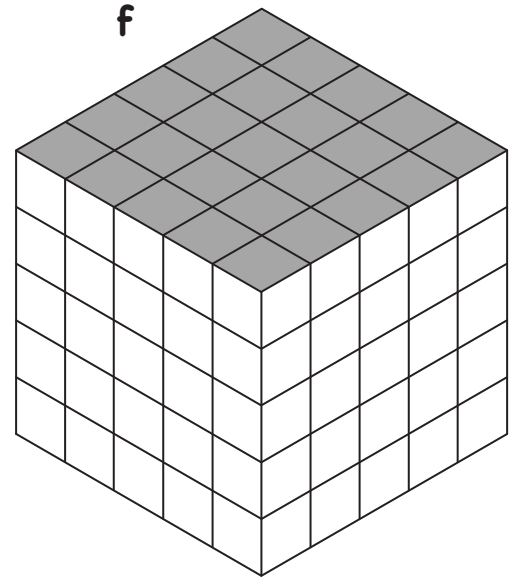
Volume = _____

e



Volume = _____

f



Volume = _____

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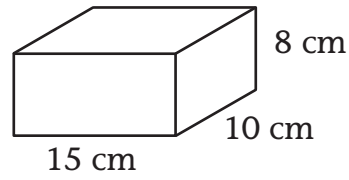
4.3 Volume of a rectangular prism B

To find the volume of a rectangular prism, first find the area of the base.
(Multiply length × width.)
Then multiply by the height.
Volume = length × width × height

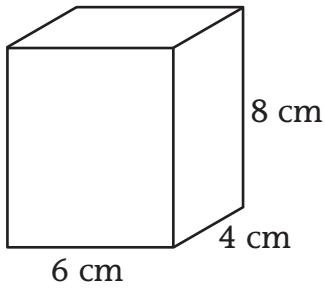
EXAMPLE



The volume of this rectangular prism is:
15 × 10 × 8
= 1 200 cm^3

1 Use the rule **volume = length × width × height** to calculate the volume of each rectangular prism.

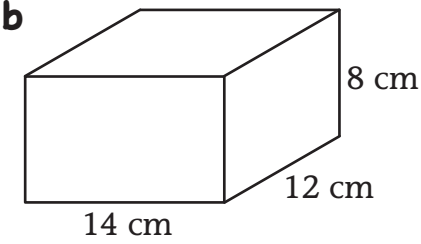
a



Volume = _____

= _____

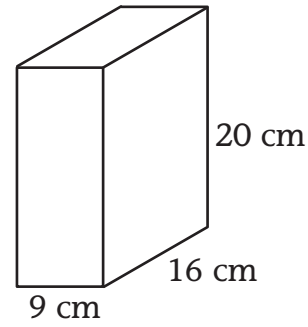
b



Volume = _____

= _____

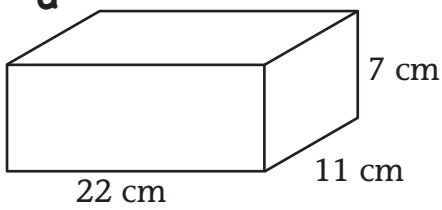
c



Volume = _____

= _____

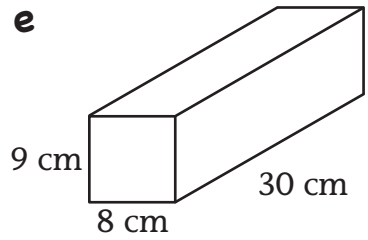
d



Volume = _____

= _____

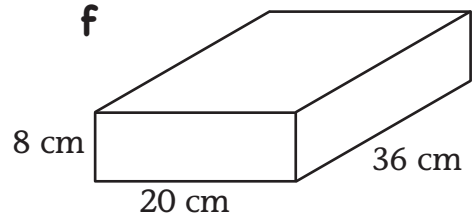
e



Volume = _____

= _____

f



Volume = _____

= _____

Answer the following questions in your workbook.

2 A swimming pool is the shape of a rectangular prism. It is 25 metres long, 8 metres wide and 2 metres deep. Calculate its volume.

3 A rectangular water storage tank measures 3.5 m by 1.4 m by 2.8 m. Calculate its volume.

4 The internal dimensions of a refrigerator are: height = 180 cm, width = 74 cm and depth = 60 cm. Calculate its volume.

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4.4 Volume of a prism

A box of chocolates is the shape of a **triangular prism**.

The triangular ends each have an area of 140 cm^2.

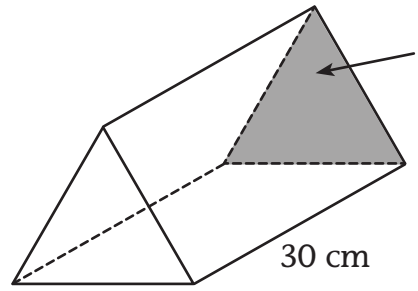
The box is 30 cm long.

The **volume of a prism** can be calculated using the rule:

Volume = area of face × height

$V = A \times h$

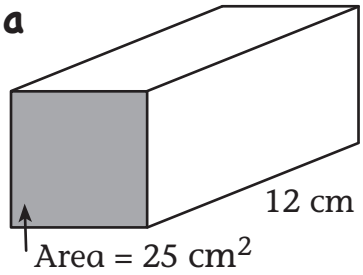
EXAMPLE



Volume
= 140 × 30
= 4 200 cm^3

1 Use the rule $V = A \times h$ to calculate the volume of each prism.

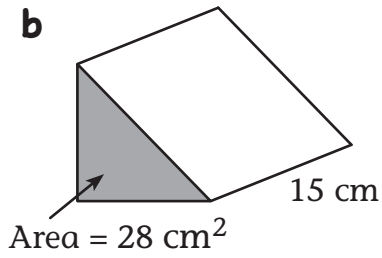
a



Volume = _____

= _____

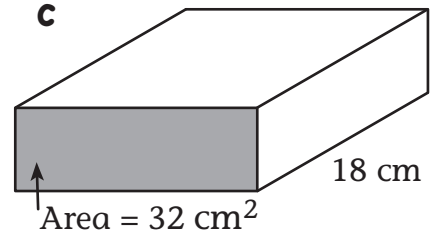
b



Volume = _____

= _____

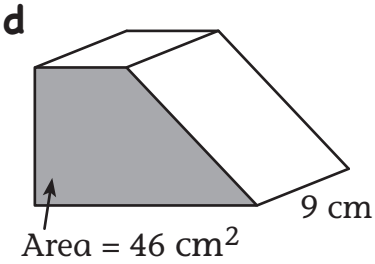
c



Volume = _____

= _____

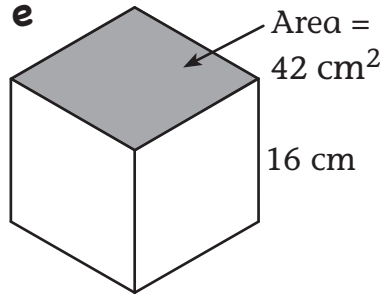
d



Volume = _____

= _____

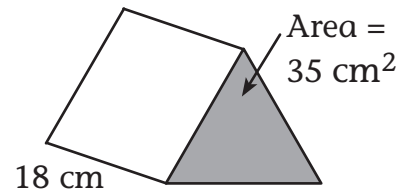
e



Volume = _____

= _____

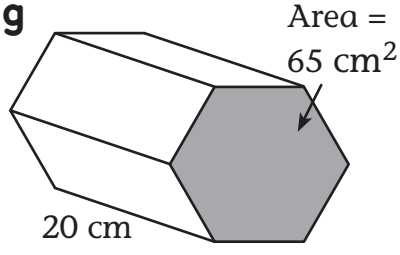
f



Volume = _____

= _____

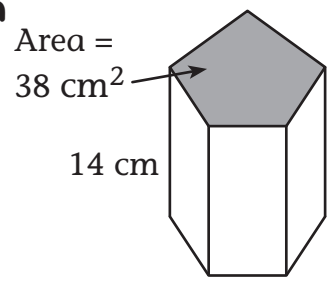
g



Volume = _____

= _____

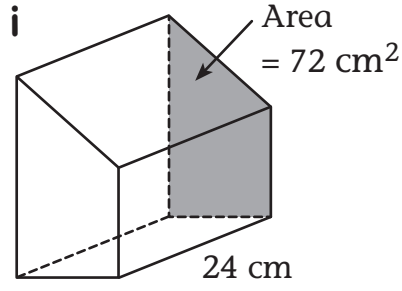
h



Volume = _____

= _____

i



Volume = _____

= _____

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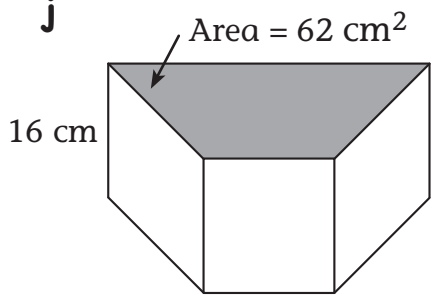
4.4 Volume of a prism

CONTINUED

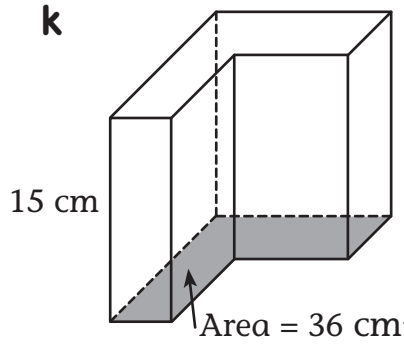
j



Volume = _____

= _____

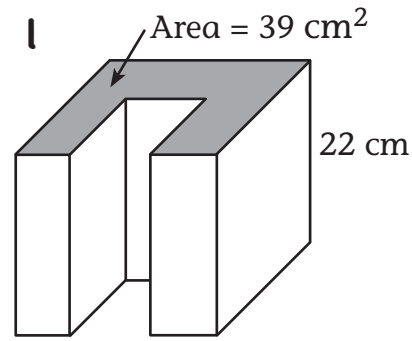
k



Volume = _____

= _____

l

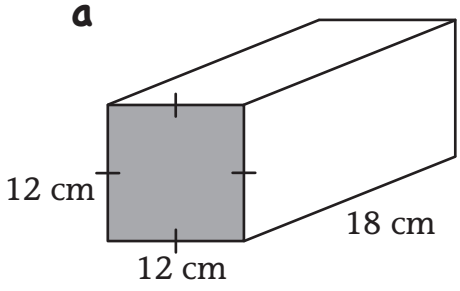


Volume = _____

= _____

2 For each prism, first calculate the area of the cross-section (the shaded face). Then calculate the volume of the prism.

a



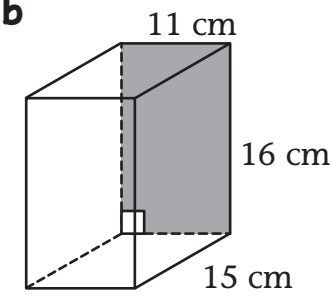
Area = _____

= _____

Volume = _____

= _____

b



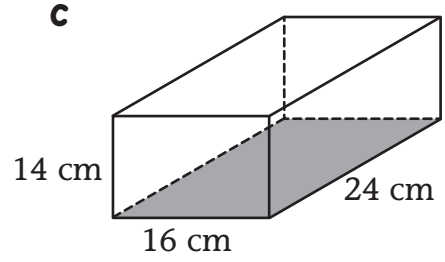
Area = _____

= _____

Volume = _____

= _____

c



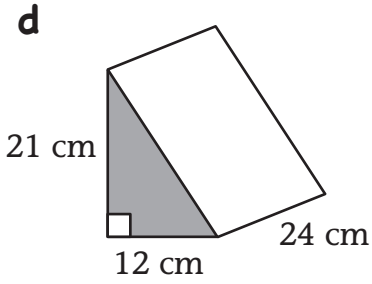
Area = _____

= _____

Volume = _____

= _____

d



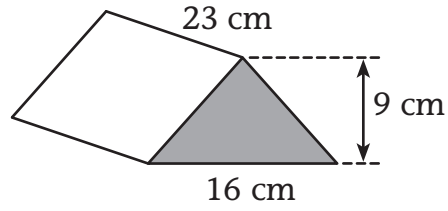
Area = _____

= _____

Volume = _____

= _____

e



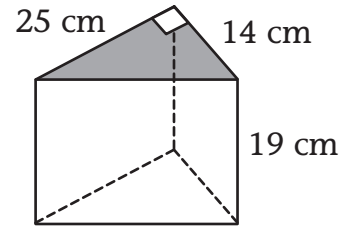
Area = _____

= _____

Volume = _____

= _____

f



Area = _____

= _____

Volume = _____

= _____

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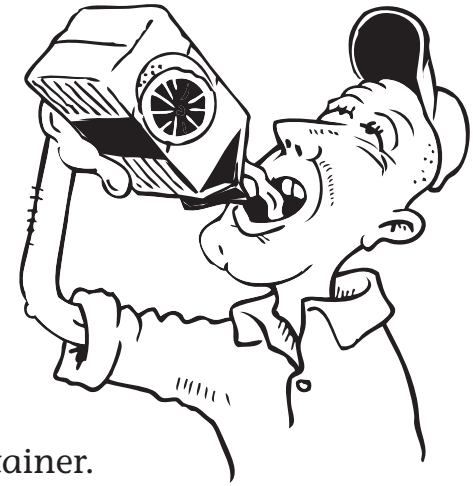
4.5 Measuring capacity

The amount of liquid or gas that a container can hold is called the **capacity** of the container.

The basic unit used to measure capacity is the **litre (L)**.

We buy soft drink, milk and petrol by the litre. A tall standard carton of milk contains 1 litre.

A cube with edges 10 cm long will hold exactly 1 litre of water.



1 Write the capacity which best matches each container. Choose from: 2 L, 7 L, 70 L, 180 L.

a a bucket of water _____

b a bottle of soft drink _____

c a bath tub _____

d a car petrol tank _____

Other common units for capacity are a **millilitre (mL)** and a **kilolitre (kL)**.

- A teaspoon holds about 5 millilitres.
- A cube with edges 1 metre long will hold exactly 1 kilolitre.

1 000 millilitres (mL) = 1 litre (L)
1 000 litres (L) = 1 kilolitre (kL)

2 Write the capacity which best matches each container. Choose from: 2 L, 5 mL, 25 mL, 60 kL, 200 mL, 375 mL, 400 L, 500 mL.

a a can of soft drink _____

b a medicine glass _____

c a teaspoon _____

d a large bottle of milk _____

e a small rain water tank _____

f a home swimming pool _____

g a bottle of tomato sauce _____

h a small glass of water _____

3 Which units – **millilitres, litres or kilolitres** – do you think are most suitable for measuring the capacity of:

a a jug of cordial? _____

b a classroom? _____

c a glass of water? _____

d an Olympic swimming pool? _____

e a garbage bin? _____

f a washing machine? _____

g a bottle of perfume? _____

h a tin of house paint? _____

i a bath tub? _____

j a road tanker? _____

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4.6 Units of capacity

How many 1 litre bottles can Amelia fill using 1.5 kilolitres of water?

Amelia needs to know many litres are equal to 1.5 kilolitres.

1 000 millilitres (mL) = 1 litre (L)
1 000 litres (L) = 1 kilolitre (kL)

Converting to smaller units:

- To convert ***kilolitres*** to ***litres***, **multiply** by 1 000.
- To convert ***litres*** to ***millilitres***, **multiply** by 1 000.

EXAMPLES

Convert 3 L to millilitres.

3 × 1 000 = 3 000 mL

Convert 1.5 kL to litres.

1.5 × 1 000 = 1 500 L

A quick way to **multiply** by **1 000**:
Move the decimal point 3 places to the **right**.

1 Convert to millilitres (mL).

a 4 L = _____ **b** 14 L = _____ **c** 1.6 L = _____

d 1.25 L = _____ **e** 0.8 L = _____ **f** 0.95 L = _____

2 Convert to litres (L).

a 3 kL = _____ **b** 50 kL = _____ **c** 4.4 kL = _____

d 12.5 kL = _____ **e** 1.25 kL = _____ **f** 0.85 kL = _____

Converting to larger units:

- To convert ***millilitres*** to ***litres***, **divide** by 1 000.
- To convert ***litres*** to ***kilolitres***, **divide** by 1 000

EXAMPLES

Convert 6 000 mL to litres.

6 000 ÷ 1 000 = 6 L

Convert 2 600 L to kilolitres.

2 600 ÷ 1 000 = 2.6 kL

A quick way to **divide** by **1 000**:
Move the decimal point 3 places to the **left**.

3 Convert to litres (L).

a 3 000 mL = _____ **b** 1 300 mL = _____ **c** 2 250 mL = _____

d 12 500 mL = _____ **e** 600 mL = _____ **f** 750 mL = _____

4 Convert to kilolitres (kL).

a 4 000 L = _____ **b** 2 500 L = _____ **c** 6 250 L = _____

d 12 000 L = _____ **e** 15 600 L = _____ **f** 950 L = _____

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4.7 Calculations with capacity

1 Find the total capacity, **in litres**, of:

a eight 7 L buckets of water.

b five 1.25 L bottles of soft drink.

c six 500 mL bottles of tomato sauce.

d twenty 375 mL cans of drink.

2 Toni poured 200 mL from a 1.25 L bottle of soft drink. How many millilitres are left in the bottle?

3 An urn holds 10 litres of water. A teapot holding 710 mL is filled from the urn. How many litres of water remain in the urn?

4 A jug holds 2 litres of water. How many 250 mL glasses could be filled from this jug?

5 A bottle holds 150 mL of medicine. Taryn was told to take 10 mL twice a day. For how many days can Taryn take this medicine before the bottle is empty?

6 Simon has some plastic cups which each hold 210 mL of drink. How many ***full*** cups of drink could be filled from three 2 litre bottles of soft drink?

7 Which has the greater capacity:

a 4 cans of lemonade which contain 375 mL each ***or*** a 2 litre bottle of lemonade?

b 6 bottles each containing 450 mL of sauce ***or*** a 2.5 litre tin of sauce?

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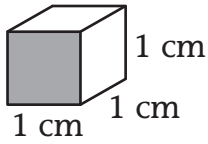
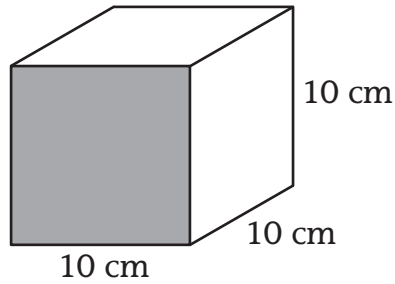
4.8 Capacity and volume A

Volume refers to the **amount of space** occupied by a solid (or three-dimensional) shape.

Capacity refers to the **amount of liquid or gas** that would fill that space.



A hollow cube with a volume of 1 cm^3 holds 1 mL of liquid.



A hollow cube with a volume of 1 000 cm^3 holds 1 litre (L) of liquid.

To convert between ***volume*** and ***capacity***:

- 1 cm^3 = 1 mL
- 1 000 cm^3 = 1 litre
- 1 m^3 = 1 000 L (or 1 kilolitre)

1 Using **1 cm^3 = 1 mL**, complete these conversions.

a 400 cm^3 = *400* mL **b** 320 cm^3 = _____ mL **c** 1 250 cm^3 = _____ mL

d 850 mL = *850* cm^3 **e** 95 mL = _____ cm^3 **f** 375 mL = _____ cm^3

2 Using **1 000 cm^3 = 1 L**, complete these conversions.

a 7 000 cm^3 = _____ L **b** 3 200 cm^3 = _____ L **c** 650 cm^3 = _____ L

d 3 L = _____ cm^3 **e** 8.4 L = _____ cm^3 **f** 0.25 L = _____ cm^3

3 Using **1 m^3 = 1 000 L**, complete these conversions.

a 6 m^3 = _____ L **b** 3.9 m^3 = _____ L **c** 0.7 m^3 = _____ L

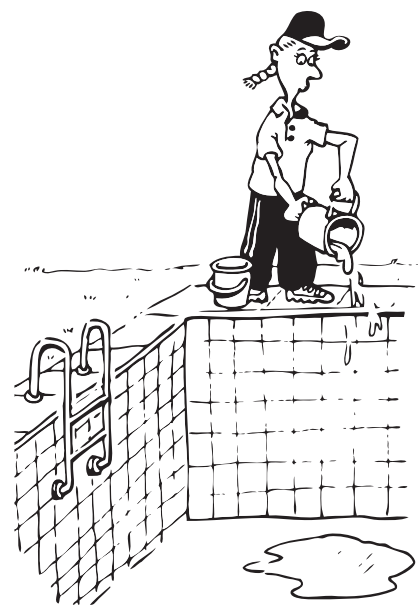
d 5 000 L = _____ m^3 **e** 2 600 L = _____ m^3 **f** 850 L = _____ m^3

Answer the following questions in your workbook.

4 The petrol tank in Duncan's car has a volume of 68 000 cm^3. How many litres of petrol will it hold?

5 A rainwater tank has a volume of 2.85 m^3. Find its capacity in litres.

6 A swimming pool holds 35 000 litres of water when full. What is its volume in cubic metres?



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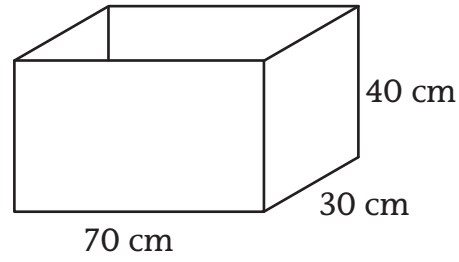
4.9 Capacity and volume B

Kym bought a fish tank in the shape of a rectangular prism, 70 cm long, 40 cm high and 30 cm wide.

Kym wanted to calculate how many litres of water it will hold.



To calculate the capacity of the tank:

Step 1: **Calculate the volume.**

V = length × width × height
= 70 × 30 × 40
= 84 000 cm^3

Step 2: **Convert to capacity.**

Using 1 cm^3 = 1 mL

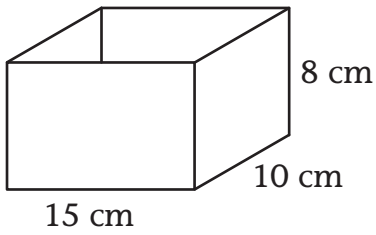
Or 1 m^3 = 1 000 L (or 1 kL)

Capacity = 84 000 mL
= **84 L**

(*Note:* We usually ignore the thickness of the sides of the container.)

1 Calculate the capacity of these containers. Give your answers **in millilitres**.

a

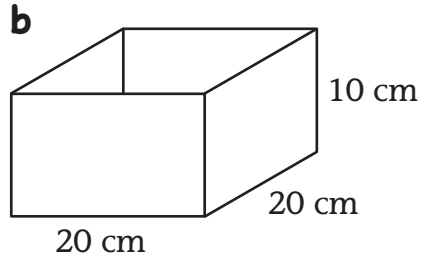


Volume = _____

= _____

Capacity = _____

b

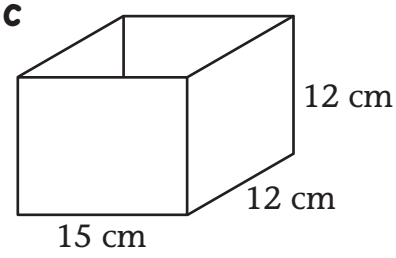


Volume = _____

= _____

Capacity = _____

c



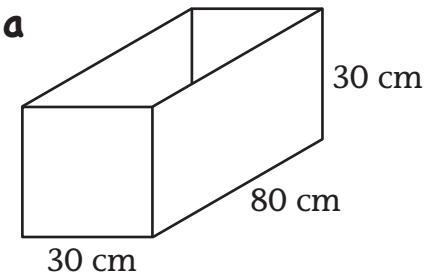
Volume = _____

= _____

Capacity = _____

2 Calculate the capacity of these containers. Give your answers **in litres**.

a

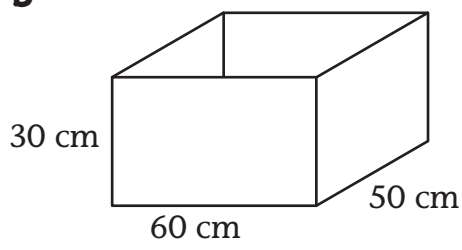


Volume = _____

= _____

Capacity = _____

b

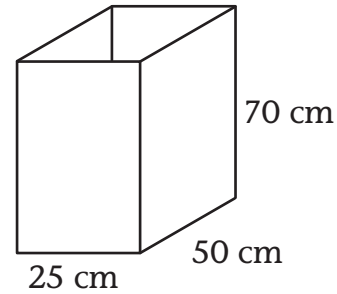


Volume = _____

= _____

Capacity = _____

c



Volume = _____

= _____

Capacity = _____

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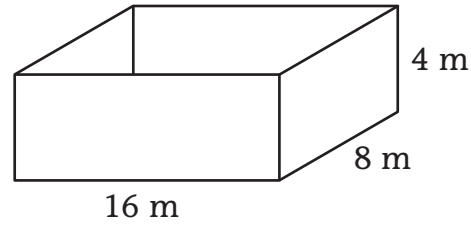
4.9 Capacity and volume B

CONTINUED

3 A rectangular water storage tank measures 16 metres by 8 metres by 4 metres. Calculate:



a the volume. _____

_____

b the capacity in litres. _____

c the capacity in kilolitres. _____

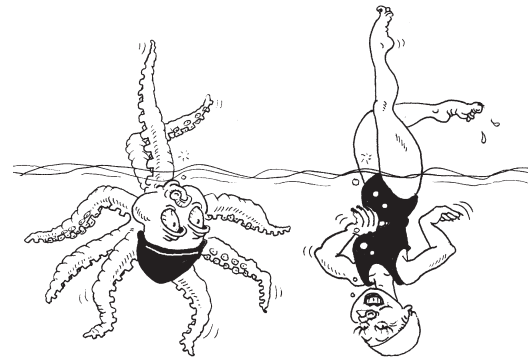
Answer the following questions in your workbook

4 A lunch box is the shape of a rectangular prism. Its dimensions are 25 cm, 15 cm and 9 cm.

a Calculate the volume of the box in cubic centimetres.

b How many millilitres of water would the lunch box hold?

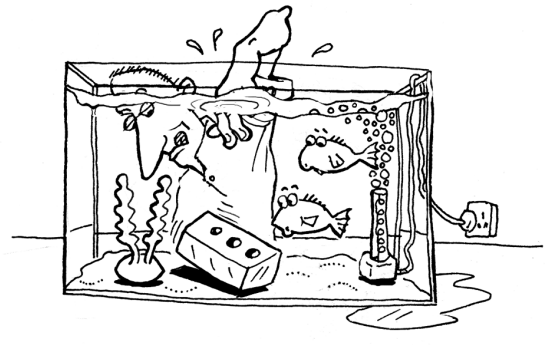
5 Gemma's swimming pool is the shape of a rectangular prism. It is 10 metres long, 5 metres wide and 1.6 metres deep.



a Calculate the volume of the swimming pool in cubic metres.

b How many litres of water are needed to fill this pool?

6 The dimensions of a rectangular fish tank are length = 70 cm, width = 50 cm and height = 30 cm.



a Calculate the volume of the tank in cubic centimetres.

b Calculate the capacity of the fish tank in litres.

7 The internal dimensions of a refrigerator are:
height = 160 cm
width = 65 cm
depth = 50 cm.

a Calculate the internal volume of this refrigerator in cm^3.

b Calculate its capacity in litres.

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Teacher's notes

OBJECTIVES

- Read and record time using digital and analogue notations.
- Convert between units of time.
- Convert between 24 hour time and 12 hour time, including am/pm notation.
- Calculate the time elapsed between starting and finishing times.
- Perform various calculations involving time.
- Read and interpret simple bus and train timetables.
- Compare the local time in various time zones in Australia.

PRIOR KNOWLEDGE

Students should be able to read analogue and digital clocks to the minute. They should be able to perform the basic operations with whole numbers, fractions and decimals using mental arithmetic or a calculator.

BACKGROUND

In the modern digital age, students are more familiar with digital time than the analogue clock.

The abbreviation *am* stands for the Latin words *ante meridiem* meaning 'before midday'. The abbreviation *pm* stands for *post meridiem* meaning 'after midday'.

Midday and midnight should not be expressed in am or pm form. '12 noon' or '12 midday' and '12 midnight' should be used, even though 12:00 pm and 12:00 am are sometimes seen.

NOTE: Different calculators might need different sequences of key presses when working with time. Check that students' calculators follow the sequence shown on these activity sheets.

STARTER ACTIVITY

Discuss the movements of the hands on an analogue clock. For example, ask: *How many minutes does it take the **minute** hand to move from one numeral to the next? How many minutes does it take the **hour** hand to move from one numeral to the next?*

Encourage students to associate the numerals 3, 6 and 9 on the face of an analogue clock with 15, 30 and 45 minutes respectively. Discuss the use of the terms *quarter past*, *half past* and *quarter to*.

ACTIVITY SHEETS

5.1 Measuring time

This activity sheet develops students' skills in converting between different units for measuring time.

Answers

1 **a** 48 h **b** 120 h **c** 36 h **d** 3 h **e** 2 h **f** 5 h
2 **a** 180 min **b** 300 min **c** 150 min **d** 2 min **e** 3 min **f** 9 min
3 **a** 180 s **b** 120 s **c** 240 s **d** 90 s **e** 210 s **f** 168 s
4 **a** 120 **b** 300 **c** 4 **d** 6 **e** 72 **f** 2 **g** 30 **h** 90 **i** 30
5 **a** 31 **b** 30 **c** 31 **d** 30 **6** **a** 4 **b** 7
7 **a** 14 **b** 50 **c** 51 **d** 123
8 **a** 3 600 **b** 1 440 **c** 168
9 **a** 4 h 30 min **b** 6 h 24 min **c** 3 h 54 min **d** 5 h 12 min **e** 6 h 15 min **f** 8 h 45 min
10 **a** 1 h 40 min **b** 2 h 30 min **c** 3 h 20 min **d** 3 h 15 min **e** 3 h 45 min **f** 5 h 25 min

5.2 The 12 hour clock

This activity sheet reminds students how to read an analogue clock and the different ways the time can be written using numerals and words.

Answers

1 **a** 8o'clock **b** 10:15 *or* a quarter past 10 **c** 3:30 *or* half past 3 **d** 6:50 *or* 10 minutes to 7
2 **a** 9:00 **b** 5:10 **c** 7:30 **d** 2:15 **e** 9:40 **f** 3:20 **g** 3:45 **h** 10:55
3 **a** 2:00 am **b** 5:00 pm **c** 6:30 am **d** 1.10 pm

5.3 24 hour time

This activity sheet develops students' skills in converting between 24 hour time and 12 hour time.

Answers

24 h time	12 h time
0300 hours	3:00 am
0400 hours	4:00 am
0500 hours	5:00 am
0600 hours	6:00 am
0700 hours	7:00 am
0800 hours	8:00 am
0900 hours	9:00 am
1000 hours	10:00 am

24 h time	12 h time
1500 hours	3:00 pm
1600 hours	4:00 pm
1700 hours	5:00 pm
1800 hours	6:00 pm
1900 hours	7:00 pm
2000 hours	8:00 pm
2100 hours	9:00 pm
2200 hours	10:00 pm

2 **a** 20 **b** 7, 7:45 am **c** 15 minutes after 11 o'clock, 11:15 am **d** 30, 1:30 pm **e** 25 minutes after 5 o'clock, 5:25 pm **f** 35 minutes after 8 o'clock, 8:35 pm
3 **a** 4:15 am **b** 6:30 am **c** 3:30 pm **d** 1:40 pm **e** 7:20 pm **f** 8:45 pm **g** 8:05 am **h** 11:15 am **i** 6:55 pm **j** 11:35 pm **k** 1:10 am **l** 10:25 am

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Teacher's notes

CONTINUED

m 12:12 am **n** 12:05 pm **o** 10:20 pm

4 **a** 0800 h **b** 0215 h **c** 1400 h **d** 1830 h
e 0920 h **f** 2040 h **g** 1200 h **h** 2330 h
i 1730 h **j** 1315 h **k** 1150 h **l** 1245 h
m 0655 h **n** 2235 h **o** 0005 h

5.4 A few minutes more

This activity sheet develops students' skills in calculating the time after a given interval. It concludes with word problems which demonstrate situations where these calculations can be used. Students could check their answers using calculators.

Answers

1 **a** 3:25 pm **b** 9:55 am **c** 8:40 am **d** 10:00 am
e 9:10 pm **f** 5:10 pm **g** 7:20 am **h** 3:15 pm

2 **a** 0430 h **b** 1525 h **c** 1150 h **d** 1855 h
e 1500 h **f** 1705 h **g** 1805 h **h** 0740 h

3 **a** 1525 h **b** 1910 h **c** 1355 h **d** 2257 h
e 1705 h **f** 2308 h

4 9:15 am **5** 11:15 am **6** 1305 h

5.5 What's the difference?

This activity sheet develops students' skills in calculating the time elapses between two given times. It concludes with word problems which demonstrate situations where these calculations can be used.

Answers

1 **a** 7 **b** 3 **c** 8 **d** 10 **e** 6 **f** 15

2 **a** 25 **b** 45 **c** 45 **d** 50 **e** 65 **f** 70

3 **a** 1 h 15 min **b** 5 h 30 min **c** 6 h 2 min
d 7 h 6 min **e** 4 h 11 min **f** 2 h 50 min
g 8 h 20 min **h** 7 h 38 min **i** 7 h 43 min
j 16 h 45 min

4 1 h 25 min **5** 2 h 50 min **6** 4 h 16 min

7 95 min

5.6 Calculations with time

This activity sheet develops students' computation skills (addition and subtraction) with time. The first page could be done without a calculator. A scientific calculator will be needed for the second page.

Answers

1 **a** 5 h 55 min **b** 6 h 10 min **c** 8 h 15 min
d 8 h 8 min **e** 9 h 57 min **f** 7 h 5 min

2 **a** 2 h 15 min **b** 2 h 40 min **c** 2 h 40 min
d 1 h 35 min **e** 3 h 30 min **f** 2 h 37 min

3 **a** 5 h 55 min **b** 6 h 18 min **c** 9 h 6 min
d 8 h 15 min

4 **a** 10:47 am **b** 11:32 pm **c** 1:37 pm
d 4:25 am **e** 2045 h **f** 1928 h

5 **a** 2 h 23 min **b** 1 h 51 min **c** 1 h 39 min
d 1 h 36 min

6 **a** 6 h 18 min **b** 5 h 15 min **c** 4 h 35 min
d 8 h 34 min **e** 7 h 45 min **f** 7 h 27 min

5.7 Catching a bus

This activity sheet develops students' skills in reading and interpreting a bus timetable shown in 12 hour time.

Answers

1 **a** 8:01 am **b** 8:40 am **c** 8:52 am

2 **a** 8 min **b** 39 min **c** 1 h 7 min (or 67 min)

3 7:14 am **4** 8:35 am **5** 8:34 am

5.8 Catching a train

This activity sheet develops students' skills in reading and interpreting a train timetable shown in 24 hour time.

Answers

1 **a** 11:11 am **b** 11:48 am **c** 12:23 pm

2 **a** 1631 h **b** 1647 h **c** 1723 h **3** 49 min

4 **a** 21 min **b** 56 min **c** 1 h 46 min

5 **a** 1 h 12 min **b** 1 h 16 min

6 The 9:11 am train **7** 11:29 am

5.9 Australian time zones

This activity sheet looks at the difference in time between the three time zones in Australia. (Daylight saving is ***not*** considered.) Students will determine the time in various cities at various times of the day.

Answers

1 **a** 9 pm **b** 8:30 pm **c** 7 pm **d** 9 pm
e 8:30 pm **f** 7 pm

2 **a** 11 am **b** 10:30 am **c** 9 am **d** 11 am
e 9 am **f** 10:30 am

3 **a** 12 noon **b** 12 noon **c** 12 noon
d 11:30 am **e** 11:30 am **f** 10 am

4 **a** 6 pm **b** 6 pm **c** 5:30 pm **d** 5:30 pm
e 4 pm **f** 6 pm

5 **a** 6:30 pm **b** 6:30 pm **c** 4:30 pm
d 6:30 pm **e** 6 pm **f** 4:30 pm

6 **a** 12:30 pm **b** 12:30 pm **c** 12:30 pm
d 10:30 am **e** 10:30 am **f** 12 noon

7 **a** 2 pm **b** 5 am **c** 8:45 am **d** 11:15 am
e 9:30 pm

EXTENSION IDEAS

Ask students to plan a journey involving a number of connections and using different modes of transport. (Use the internet to obtain the timetables.)

Compare times at major cities around the world and calculate the differences between time zones. Many diaries and websites provide world time information.

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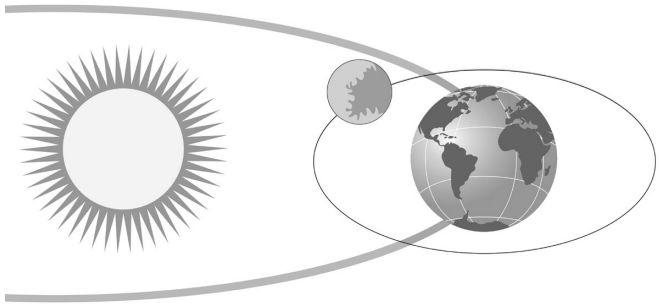
5.1 Measuring time

People in ancient times did not have clocks, watches or mobile phones to tell the time. Their time was measured against the rise and fall of the Sun.

One **day** is the time taken by the Earth to spin once on its own axis.

One **year** is the time taken by the Earth to complete one orbit around the Sun.



Units of time

1 week = 7 days
1 day = 24 hours (h)
1 hour = 60 minutes (min)
1 minute = 60 seconds (s)

One year contains $365\frac{1}{4}$ days.

There are 365 days in a normal calendar year.

Every fourth calendar year is called a leap year. A leap year contains 366 days.

A year is divided into 12 calendar months. The number of days in a month is not fixed.

1 How many hours are equal to:

a 2 days? *2 × 24 = 48 h*

b 5 days? _____

c $1\frac{1}{2}$ days? _____

d 180 min? *180 ÷ 60 = 3 h*

e 120 min? _____

f 300 min? _____

2 How many minutes are equal to:

a 3 hours? *3 × 60 = 180 min*

b 5 hours? _____

c $2\frac{1}{2}$ hours? _____

d 120 s? *120 ÷ 60 = 2 min*

e 180 s? _____

f 540 s? _____

3 How many seconds are equal to

a 3 min? *3 × 60 = 180 s*

b 2 min? _____

c 4 min? _____

d $1\frac{1}{2}$ min? _____

e $3\frac{1}{2}$ min? _____

f 2.8 min? _____

4 Complete.

a 2 h = _____ min

b 5 min = _____ s

c 240 min = _____ h

d 360 s = _____ min

e 3 days = _____ h

f 48 h = _____ days

g $\frac{1}{2}$ hour _____ min

h 1.5 h = _____ min

i $\frac{1}{2}$ min = _____ s

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5.1 Measuring time

CONTINUED

5 How many days are there in the following months?

a March _____ **b** June _____ **c** August _____ **d** November ___

6 How many months have:

a 30 days? _____ **b** 31 days? _____

7 How many days, inclusive, are there from:

a 16th April to 29th April? _____ **b** 29th July to 16th September?_____

c 23rd May to 12th July? _____ **d** 9th April to 9th August? _____

8 Use your calculator to work out:

a the number of seconds in an hour. _____

b the number of minutes in a day. _____

c the number of hours in 1 week. _____

Time calculations can be completed using the [° ′ ″] or [DMS] key on your calculator.

EXAMPLE

Convert 3.4 hours to hours and minutes.

Press 3.4 [° ′ ″]. This gives 3°24′0″ on the display, which means 3 h 24 min.

9 Use the [° ′ ″] key on your calculator to convert these to hours and minutes.

a 4.5 hours = _____ **b** 6.4 hours = _____ **c** 3.9 hours = _____

d 5.2 hours = _____ **e** 6.25 hours = _____ **f** 8.75 hours = _____

EXAMPLE

Convert 140 minutes to hours and minutes.

Step 1: Divide by 60 to convert to hours. 140 ÷ 60 = 2.3333 ...

Step 2: Press [° ′ ″] or [DMS] to get 2°20′0″ on the display. This means **2 h 20 min**.

10 Convert these amounts of time to hours and minutes.

a 100 min = _____ **b** 150 min = _____ **c** 200 min = _____

d 195 min = _____ **e** 225 min = _____ **f** 325 min = _____

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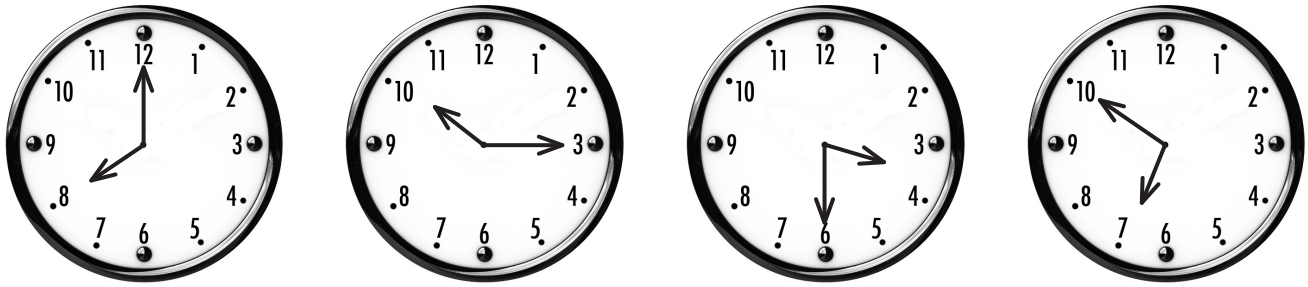
5.2 The 12 hour clock

Lyn was visiting her grandmother. She looked at a clock on the wall.
What time is shown on this clock?

The time shown on an **analogue clock** may be written in a number of ways.
The time shown on this clock is **2:25** or **twenty-five past two**.

1 Write the time shown on each clock face.

a _____ **b** _____ **c** _____ **d** _____



To write the time using words, write how many minutes (up to 30) *after* (past) or *before* (to) the hour.
The half and quarter hours are used in place of 30 minutes or 15 minutes.

EXAMPLES

1:20 is written as ***twenty past one***
7:40 is written as ***twenty to eight***
8:30 is written as ***half past eight***
3:15 is written as ***a quarter past three***
6:45 is written as ***a quarter to seven***

2 Write these times as they would appear on a digital 12-hour clock.

a nine o'clock _____
b ten past five _____
c half past seven _____
d a quarter past two _____
e twenty to ten _____
f twenty past three _____
g a quarter to four _____
h five to eleven _____

When we write digital 12-hour times, we usually add ***am*** or ***pm*** at the end.
The letters ***am*** tell you that it is morning (before noon).
The letters ***pm*** tell you that it is afternoon or evening.

3 Write these times in numerals. Use *am* or *pm* to indicate morning or afternoon.

a two o'clock in the morning *2:00 am*
b five o'clock in the afternoon _____
c half past six in the morning _____
d ten past one in the afternoon _____

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5.3 24 hour time

Holly wanted to catch a plane from Adelaide to Sydney. She checked the timetable on the internet. She saw that a plane left Adelaide at 1435.

24-hour times are used on many timetables and in some overseas countries.

24-hour time starts from midnight (0000 hours).
All times are written using 4 digits.
The first two digits tell you the number of hours after midnight.
The last two digits tell you the number of minutes.

1 The table shows equivalent hours in 24-hour and 12-hour time. Complete the missing times.

24-hour time	12-hour time
0000 hours	12:00 midnight
0100 hours	1:00 am
0200 hours	2:00 am
0300 hours	3:00 am
0400 hours	
	5:00 am
0600 hours	6:00 am
0700 hours	7:00 am
	8:00 am
0900 hours	
1000 hours	10:00 am
1100 hours	11:00 am

24-hour time	12-hour time
1200 hours	12:00 noon (midday)
1300 hours	1:00 pm
1400 hours	2:00 pm
1500 hours	
	4:00 pm
1700 hours	
1800 hours	6:00 pm
	7:00 pm
2000 hours	
2100 hours	9:00 pm
	10:00 pm
2300 hours	11:00 pm

The 24-hour time **1435** is 35 minutes after 2:00 pm.

That is **2:35 pm** in 12-hour (digital) time.

2 Complete these statements.

a 0920 hours is _____ minutes after 9 o'clock. It is 9:20 am in 12 hour time.

b 0745 hours is 45 minutes after ____ o'clock. It is _____ in 12 hour time.

c 1115 hours is _____ minutes after ____ o'clock. It is _____ in 12 hour time.

d 1330 hours is _____ minutes after 1 o'clock. It is _____ in 12 hour time.

e 1725 hours is _____ minutes after ____ o'clock. It is _____ in 12 hour time.

f 2035 hours is _____ minutes after ____ o'clock. It is _____ in 12 hour time.

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5.3 24 hour time

CONTINUED

To convert from 24-hour time to 12-hour (am/pm) time,
insert a colon (:) after the first two digits.
Then look at the first two digits.

- If they are *less than 12* then it is morning. Write ***am*** after your answer.
- If they are *12 or more* then it is afternoon or evening. Write ***pm*** after your answer.
- If it is *1300 hours or later*, subtract 12 from the first two digits.

EXAMPLES

0705 hours = 7:05 am
1030 hours = 10:30 am
1250 hours = 12:50 pm
1435 hours = 2:35 pm

3 Convert these times to 12-hour (am/pm) time.

a 0415 hours = _____ **b** 0630 hours = _____ **c** 1530 hours = _____
d 1340 hours = _____ **e** 1920 hours = _____ **f** 2045 hours = _____
g 0805 hours = _____ **h** 1115 hours = _____ **i** 1855 hours = _____
j 2335 hours = _____ **k** 0110 hours = _____ **l** 1025 hours = _____
m 0012 hours = _____ **n** 1205 hours = _____ **o** 2220 hours = _____

To convert from 12-hour (am/pm) time to 24-hour time,
remove the am/pm and the colon (:).

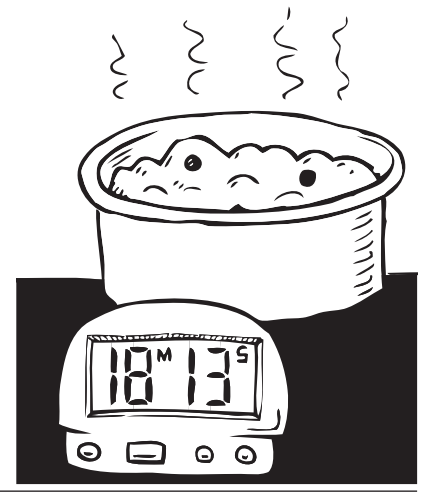
- If the time is between midnight (0000 hours) and 1 am (0100 hours), the first two digits become zeros.
- If the time was earlier than 10 am (1000 hours), add a zero in front, so there are 4 digits.
- If the time was after 1 pm, add 12 to the number of hours (the first two digits).

EXAMPLES

9:30 am = 0930 hours
8:15 am = 0815 hours
2:50 pm = 1450 hours
7:35 pm = 1935 hours

4 Convert these times to 24-hour time.

a 8:00 am = _____ **b** 2:15 am = _____
c 2:00 pm = _____ **d** 6:30 pm = _____
e 9:20 am = _____ **f** 8:40 pm = _____
g 12 noon = _____ **h** 11:30 pm = _____
i 5:30 pm = _____ **j** 1:15 pm = _____
k 11:50 am = _____ **l** 12:45 pm = _____
m 6:55 am = _____ **n** 10:35 pm = _____
o 12:05 am = _____



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5.4 A few minutes more

A train departed Bendigo at 10:30 am. It arrived at Woodend 50 minutes later.

What time did this train arrive at Woodend?

Count up the minutes to find the answer.

EXAMPLE

What is the time 50 minutes after 10:30 am?

30 minutes after 10:30 am gives 11:00 am.
We need an extra 20 minutes to give 50 minutes.
The answer is **11:20 am**.

1 What is the time:

a 15 minutes after 3:10 pm?_____ **b** 30 minutes after 9:25 am?_____

c 25 minutes after 8:15 am?_____ **d** 45 minutes after 9:15 am?_____

e 15 minutes after 8:55 pm?_____ **f** 25 minutes after 4:45 pm?_____

g 50 minutes after 6:30 am?_____ **h** 40 minutes after 2:35 pm?_____

2 What is the time:

a 10 minutes after 0420 h?_____ **b** 20 minutes after 1505 h?_____

c 15 minutes after 1135 h?_____ **d** 30 minutes after 1825 h?_____

e 45 minutes after 1415 h?_____ **f** 25 minutes after 1640 h?_____

g 50 minutes after 1715 h?_____ **h** 55 minutes after 0645 h?_____

3 What is the time:

a 1 h 5 min after 1420 h?_____ **b** 1 h 50 min after 1720 h?_____

c 2 h 15 min after 1140 h?_____ **d** 2 h 45 min after 2012 h?_____

e 3 h 25 min after 1340 h?_____ **f** 3 h 18 min after 1950 h?_____

Answer the following questions in your workbook

4 A train left Newcastle at 7:35am. It arrived at Woy Woy 1 hour and 40 minutes later. At what time did the train arrive at Woy Woy?

5 Duncan drove his car from Parramatta to Goulburn.
He left Parramatta at 8:50 am. The drive took 2 hours and 25 minutes.
At what time did Duncan arrive in Goulburn?

6 A plane departed Sydney at 0955 h. It arrived at Cairns 3 hours and 10 minutes later. At what time did the plane arrive in Cairns?

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5.5 What's the difference?

Emily caught a train from Sunbury to Melbourne.
The train departed Sunbury at 8:06 am.
It arrived at Melbourne at 8:45 am.
How long did the journey take?
Count up the minutes.

EXAMPLE

From 8:06 to 8:10 is **4 minutes.**
From 8:10 to 8:45 is **35 minutes.**
The total from 8:06 to 8:45 is **39 minutes.**

1 How many hours are there between:

a midnight and 7 am? _____
b 9 am and midday? _____
c 3 am and 11 am? _____
d 5 am and 3 pm? _____
e 10 am and 4 pm? _____
f 6 am and 9 pm? _____

2 How many minutes are there between:

a 7:15 am and 7:40 am? _____
b 2:30 pm and 3:15 pm? _____
c 8:20 am and 9:05 am? _____
d 4:15 pm and 5:05 pm? _____
e 10:45 am and 11:50 am? _____
f 11:25 am and 12:35 pm? _____

3 Calculate the length of time in hours and minutes from:

a 0834 until 0949. _____
b 1206 until 1736. _____
c 0748 until 1350. _____
d 0943 until 1649. _____
e 1930 until 2341. _____
f 0735 until 1025. _____
g 1450 until 2310. _____
h 1027 until 1805. _____
i 0952 until 1735. _____
j 0645 until 2330. _____

Answer the following questions in your workbook.

4 A train leaves Sydney's Central station at 4:45 pm. It arrived at Gosford at 6:10 pm. How long was the journey from Central to Gosford?

5 While shopping, Lee left his car in a car park from 9:35 am to 12:25 pm. For how long was his car in the car park?

6 Kaye parked her car from 11:27 am to 3:43 pm. For how long was her car in the car park?

7 A plane departed Launceston at 1320 h. It flew direct to Sydney and arrived at 1455 h. How many minutes did the flight take?

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5.6 Calculations with time

EXAMPLE

Adding time

$$\begin{array}{r} 2\text{ h }35\text{ min} \\ +\ 3\text{ h }45\text{ min} \\ \hline \mathbf{5\text{ h }80\text{ min}} \end{array} \quad \rightarrow \quad \begin{array}{r} 5\text{ h} \\ +\ 1\text{ h }20\text{ min} \\ \hline \mathbf{6\text{ h }20\text{ min}} \end{array}$$

80 min = 1 h 20 min

Remember, 60 minutes = 1 hour

1 Complete these additions.

a 3 h 40 min + 2 h 15 min

_____

b 4 h 20 min + 1 h 50 min

_____

c 3 h 30 min + 4 h 45 min

_____

d 5 h 43 min + 2 h 25 min

_____

e 6 h 18 min + 3 h 39 min

_____

f 4 h 46 min + 2 h 19 min

_____

EXAMPLE

Subtracting time

$$\begin{array}{r} 5\text{ h }15\text{ min} \\ -\ 2\text{ h }45\text{ min} \\ \hline \end{array} \quad = 4\text{ h} + 1\text{ h }15\text{ min} = 4\text{ h }75\text{ min} \quad \begin{array}{r} -\ 2\text{ h }45\text{ min} \\ \hline \mathbf{2\text{ h }30\text{ min}} \end{array}$$

When you are subtracting a larger number of minutes, exchange 1 hour for 60 minutes.

2 Complete these subtractions.

a 3 h 50 min − 1 h 35 min

_____

b 4 h 20 min − 1 h 40 min

_____

c 5 h 10 min − 2 h 30 min

_____

d 5 h 15 min − 3 h 40 min

_____

e 6 h 25 min − 2 h 55 min

_____

f 4 h 20 min − 1 h 43 min

_____

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5.6 Calculations with time

CONTINUED

Time calculations can be done on your calculator using the [° ′ ″] or [DMS] key.

EXAMPLE

3 hours 52 minutes + 2 hours 15 minutes

The calculator steps are:

[3] [° ′ ″] [5] [2] [° ′ ″] [+] [2] [° ′ ″] [1] [5] [° ′ ″] [=]

The calculator display shows 6.116666 ...
To show hours and minutes, press [SHIFT] [° ′ ″]
The answer is **6 h 7 min**.

3 Add these times together.
Give your answers in hours and minutes.

a 4 h 25 min + 1 h 30 min = _____

b 3 h 46 min + 2 h 32 min = _____

c 5 h 19 min + 3 h 47 min = _____

d 4 h 48 min + 3 h 27 min = _____

4 What time will it be:

a 2 h 2 min after 8:45 am? _____

b 1 h 40 min after 9:52 pm? _____

c 3 h 12 min after 10:25 am? _____

d 8 h 55 min after 7:30 pm? _____

e 5 h 35 min after 1510 h? _____

f 7 h 23 min after 1205 h? _____

5 Subtract these times.
Give your answers in hours and minutes.

a 3 h 55 min – 1 h 32 min = _____

b 4 h 15 min – 2 h 24 min = _____

c 4 h 25 min – 2 h 46 min = _____

d 5 h 28 min – 3 h 52 min = _____

When you want to find the difference between two times,
first convert both times to 24-hour time.
Then use the calculator's [° ′ ″] or [DMS] key.

6 What is the difference in time between:

a 3:25 am and 9:43 am? _____

b 9:10 am and 2:25 pm? _____

c 10:53 am and 3:28 pm? _____

d 11:43 am and 8:17 pm? _____

e 0728 h and 1513 h? _____

f 1315 h and 2042 h? _____

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5.7 Catching a bus

Cate lives at Chittaway. She needed to catch a bus to the nearest train station at Tuggerah. Cate looked at a copy of the local bus timetable.

Part of a timetable for buses travelling between The Entrance and Wyong is shown below. These times are for weekday mornings (Monday to Friday).

Bus number	**THE ENTRANCE Terminal**	STELLA STREET	**BAY VILLAGE**	SHERRY STREET	PINDARRI ROAD	KILKENNY PARADE	BERKELEY ROAD	CHITTAWAY ROAD	**TUGGERAH STATION**	**WYONG**
24	6:52	6:59	7:08	7:17	·····	7:25	7:33	7:41	7:49	7:58
26	7:06	7:14	7:22	·····	·····	7:38	·····	7:58	8:06	8:16
25	7:45	7:53	8:01	8:06	8:19	.....	8:24	8:34	8:40	8:52
26	7:58	8:05	8:14	·····	·····	8:26	·····	8:45	8:51	8:59
25	8:35	8:43	8:55	9:00	9:11	.....	9:15	9:25	9:30	9:41
26	9:03	9:11	9:19	·····	·····	9:30	·····	9:45	9:50	10:02
25	9:26	9:34	9:45	9:50	10:01	·····	10:05	10:15	10:20	10:31

1 At what time does the 7:45 am bus from The Entrance reach:

a Bay Village? _____ **b** Tuggerah Station? _____ **c** Wyong? _____

2 How long will it take the 7:45 am bus (number 25) from The Entrance to reach:

a Stella Street? _____ **b** Berkeley Road? _____ **c** Wyong? _____

3 Peter lives near the bus stop at Stella Street.
He catches a bus to work at Wyong. Peter starts work at 8:30 am.
What is the latest time he can catch a bus to arrive at work on time? _____

4 Leslie lives near the bus terminal at The Entrance.
She promised to meet her friend at Bay Village at 9:05 am.
What is the latest time she can catch a bus to arrive on time? _____

5 Cate needs to catch a train leaving Tuggerah Station at 8:50 am. What is the latest time she can take a bus from Chittaway Road and still catch her train?

_____

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5.8 Catching a train

Janet lives at Castlemaine. She needs to catch a train to Melbourne.
Janet found the train timetable on the internet.

Part of the timetable for trains travelling from Bendigo to Melbourne on Saturdays is shown in this table.

Service No.	8024	8026	8028	8030	8032	8034	8042
BENDIGO dep.	08:30	09:26	10:30	11:38	12:28	14:34	15:24
CASTLEMAINE	08:51	09:50	10:51	11:59	12:49	14:55	15:48
KYNETON	09:11	10:15	11:11	12:18	13:09	15:15	16:11
GISBORNE	09:29	10:37	11:29	12:34	13:27	15:33	16:31
SUNBURY	09:48	10:55	11:48	12:49	13:46	15:50	16:47
MELBOURNE arr.	10:23	11:31	12:23	13:24	14:22	16:27	17:23

1 At what time does the 10:30 train from Bendigo arrive at:

a Kyneton? _____ **b** Sunbury? _____ **c** Melbourne? _____

2 Janet caught the 15:48 train from Castlemaine.
At what time will this train arrive at:

a Gisborne? _____ **b** Sunbury? _____ **c** Melbourne? _____

3 How long will it take the 09:26 train (Service No. 8026) from Bendigo to reach Kyneton?

_____

4 How long will it take the 11:38 train (Service No. 8030) from Bendigo to reach:

a Castlemaine? _____ **b** Gisborne? _____ **c** Melbourne? _____

5 How long will it take to reach Melbourne on:

a the 09:11 train from Kyneton? _____

b the 10:15 train from Kyneton? _____

6 Which of the two trains in question 5, takes the shorter time to travel between Kyneton and Melbourne?

_____

7 Mitchell lives at Gisborne. He wants to meet a friend in Melbourne at 1 pm. What is the latest time he can catch a train from Gisborne to arrive at Melbourne before 1 pm? _____

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5.9 Australian time zones

Lesley lives in Brisbane. Her friend Nadine lives in Perth.
They regularly chat on the telephone.
Nadine must be careful when she phones Lesley at night.
It is two hours later in Brisbane than the time in Perth.

Australia is divided into three **time zones**.

The zones are shown on the map.

All places in the same time zone have the same time.

As you travel in an easterly direction →, the times increase (are ahead).

If you travel west ←, the times decrease (are behind).

Eastern Standard Time (Queensland, New South Wales, Victoria and Tasmania) is 30 minutes ahead of Central Standard Time.
It is 30 minutes earlier in Adelaide than Brisbane.

Eastern Standard Time is 2 hours ahead of Western Standard Time.
It is 2 hours earlier in Perth than Brisbane.

For example, when the time is 9 pm in Brisbane, it is 8:30 pm in Adelaide and 7 pm in Perth.

1 When it is 9 pm in Brisbane, what is the time in:

a Melbourne? _____ **b** Adelaide? _____ **c** Perth? _____

d Sydney? _____ **e** Alice Springs? _____ **f** Kalgoorlie? _____

2 When it is 11 am in Melbourne, what is the time in:

a Brisbane? _____ **b** Adelaide? _____ **c** Perth? _____

d Hobart? _____ **e** Broome? _____ **f** Darwin? _____

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5.9 Australian time zones

CONTINUED

Western Standard Time (Western Australia) is 2 hours behind Eastern Standard Time. It is 2 hours later in Sydney than Perth.

Western Standard Time is 1 hour and 30 minutes behind Central Standard Time. It is 1 hour and 30 minutes later in Adelaide than Perth.

For example, when the time is 10 am in Perth, it is 12 noon in Sydney and 11:30 am in Adelaide.

3 When it is 10 am in Perth, what is the time in:

a Brisbane? _____ **b** Melbourne? _____ **c** Hobart? _____

d Alice Springs? _____ **e** Darwin? _____ **f** Broome? _____

4 When it is 4 pm in Kalgoorlie, what is the time in:

a Hobart? _____ **b** Sydney? _____ **c** Adelaide? _____

d Darwin? _____ **e** Perth? _____ **f** Canberra? _____

Central Standard Time (South Australia and Northern Territory) is 30 minutes behind Eastern Standard Time. It is 30 minutes later in Melbourne than Adelaide.

Central Standard Time is 1 hour and 30 minutes ahead of Western Standard Time. It is 1 hour and 30 minutes earlier in Perth than Adelaide.

For example, when the time is 6 pm in Adelaide, it is 6:30 pm in Hobart and 4:30 pm in Perth.

5 When it is 6 pm in Adelaide, what is the time in:

a Sydney? _____ **b** Melbourne? _____ **c** Perth? _____

d Cairns? _____ **e** Alice Springs? _____ **f** Kalgoorlie? _____

6 When it is 12 noon in Darwin, what is the time in:

a Canberra? _____ **b** Brisbane? _____ **c** Cairns? _____

d Perth? _____ **e** Kalgoorlie? _____ **f** Alice Springs? _____

7 What is the time in:

a Perth, when it is 4 pm in Melbourne? _____

b Adelaide, when it is 5:30 am in Brisbane? _____

c Alice Springs, when it is 9:15 am in Canberra? _____

d Brisbane, when it is 10:45 am in Darwin? _____

e Hobart, when it is 7:30 pm in Perth? _____

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